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A systematic review on the use of patient-held health records in low and middle-income countries.

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A systematic review on the use of patient-held health records in low and middle-income countries.

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ABSTRACT

Objective

To review the available evidence on the benefit of PHRs, other than maternal and child health records for improving the availability of medical information for handover communication between healthcare providers (HCPs) and/or between HCPs and patients in LMICs.

Methods

The literature searches were conducted in PubMed, EMBASE, CINAHL databases for manuscripts without any restrictions on dates/language. Additionally articles were located through citation checking using previous systematic reviews and a grey literature search by contacting experts, searching websites of World Health Organisation website and Google scholar.

Results

Six observational studies in four LMICs met the inclusion criteria. However, no studies reported on health outcomes after using PHRs. Studies in the review reported patients' experience of carrying the records to HCPs (n=3), quality of information available to HCPs (n=1), and the utility of these records to patients (n=6) and HCPs (n=4). Most patients carry PHRs to HCP visits. One study assessed the completeness of clinical handover information and found that only 41% (161/395) of PHRs were complete with respect to key information on diagnosis, treatment and follow-up. No protocols or guidelines for HCPs were reported for use of PHRs. The HCPs perceived the use of PHRs improved medical information availability from other HCPs. From the patient perspective, PHRs functioned as documented source of information about their own condition.

Conclusion: Limited data on existing PHRs make their benefits for improving health outcomes in LMICs uncertain. This knowledge gap calls for research on understanding the dynamics and outcomes of PHR use by patients and HCPs and in health systems interventions.

Keywords -**International health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH**

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Study design that enabled to accumulate and present current evidence on use of patient-held health records other than maternal and child health records in low and middle income countries.
- Thorough literature search of three major electronic databases and reporting as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.
- Inconclusive results due to no evidence from trials.
- Small number of studies without reporting on health outcomes, making it difficult to determine usefulness for patients with non-communicable diseases.

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INTRODUCTION

To meet sustainable development goals[1] and deliver quality care[2], health systems in low and middle income countries (LMICs) need to be redesigned and strengthened.[3] In line with this agenda, the World Health Organization (WHO) has formulated a health systems framework that describes six building blocks for health systems; of these, good service delivery comprises the delivery of effective, safe, quality health care and continuity of care is an important characteristic for delivering quality care.[4]

In many LMICs, healthcare delivery is usually organised around acute and single episode care, from different facilities and health care providers (HCP).[5] Such a model of healthcare delivery is inadequate due to increasing non-communicable disease (NCD) burden in LMICs.[6] Continuity of care needed for efficient and effective care, is simply defined as "the seamless provision of healthcare between settings and over time".[7] Informational continuity represents the cornerstone of continuity of care and is defined as the use of information on past events and personal circumstances to make current care appropriate for each individual.[8] Handover communication not only relates to the process of passing patient-specific health information between HCPs, from one visit to another, but also from HCPs to the patient and family for ensuring patient care-continuity and self-management.[9]

Complete and accurate clinical information about previous management and treatment is necessary for managing all patients but especially patients with chronic conditions due to a need for on-going care from a range of HCPs.[10] Inadequate documented handover for HCPs often results in adverse events[11 12] or increased in-patient visits[13], repeated tests or examinations leading to costly healthcare, burdensome to patients and waste of resources.[13] The ability to view patients' medical records, across healthcare visits and facilities helps to facilitate health information exchange.[14] The absence of out-patient medical records,[15 16]difficulties in retrieval of facility based records for out-

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patient visits,[17] and lack of integrated patient electronic medical records between departments and between health facilities in LMICs contribute to poor informational continuity for HCPs.[18]

A patient-held health record (PHR) can be a viable solution for improving sharing of documented medical information across HCPs.[19] For the purpose of this review, PHRs are formal records given to patients in the form of booklets, which contain patient's medical information. Our hypothesis is that such PHRs with documented medical notes from HCPs can be used as a tool for improving medical information availability for HCPs.[15 20]

Paper-based PHRs are used extensively in maternity care, as home-based records or maternal and child health handbooks for sharing information across antenatal, labour and post-natal care by all HCPs even in LMIC settings.[21] Three systematic reviews on maternity and child records indicate that PHRs improve HCP-patient communication and facilitate referrals, and both HCPs and women give positive feedback on their use.[22-25]

There are several systematic reviews[19 26-29] of PHRs for chronic conditions from high income countries (HIC) which were evaluated for usefulness in cancer care,[28] and chronic conditions,[26 27] and for people with severe mental illness[29] or undocumented immigrants.[19] These condition-specific PHRs were designed to mainly improve communication between HCP and patients.[30] The reviews found that PHRs may improve a patient's sense of control and empowerment, leading to better involvement in their own care.[28 29]

There is no systematic review of PHRs for conditions other than maternal and child health from LMICs, while there is evidence that generic PHRs do exist in some LMICs.[31] A study in India found that patients were carrying unstructured sheets of papers to out-patient visits and patient-held health booklets were an acceptable and pragmatic intervention to improve information exchange for chronic NCD patients by patients and HCPs.[15]

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This review aimed to investigate the benefits of generic PHRs used in LMICs to ensure the availability of medical information for clinical handover communication and continuity of care between HCPs or outpatient health facility visits, with or without their use for communication between HCPs and patients. This review investigated non-maternal-child health PHRs, designed to record patient histories and healthcare information across a range of conditions and HCPs, including primary and secondary care and not specific to one facility or one speciality.

METHODS

The protocol for this review is registered with the PROSPERO International Prospective Register of Systematic Reviews (CRD42019139365)[32].

Selection criteria

The published literature was searched for quantitative randomised controlled trials (RCTs), cluster RCTs, quasi-experimental studies, cross sectional and pre-post study designs, qualitative and mixed-method studies. Case reports, commentaries and review articles were excluded studies. The eligibility criteria are provided in Table 1.

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Table 1 Inclusion and exclusion criteria

Title	Inclusion criteria	Exclusion criteria
Population	• Participants (patients, carers and/or health care HCPs) from any health background, of 18 years or more, using a PHR were considered. • All healthcare settings within LMICs (as defined by the World Bank 2018)[33] were considered.	• Children and pregnant women • Patients/ carers/HCPs from HIC
Intervention	• Studies which include design, implementation or evaluation of formalised PHRs , with the purpose of improving information exchange and communication between visits to the same facility for HCPs, across different health care facilities (primary to secondary/vice versa), and documented information from HCPs to patients(for patients 'own care). • Studies which describe PHRs which are clinically focused and person specific or generic PHRs. Studies which describe condition specific PHRs, they should include NCDs such as Diabetes Mellitus, hypertension, cardiovascular diseases, chronic respiratory diseases or cancer. • PHRs are defined as any formal medical document in the form of a booklet held by patients, which can be used across health care settings or visits, contain patient histories and healthcare information to guide healthcare workers providing care.	• Facility based medical records which are for HCP use. • Home based records used for maternal and child health care. • Patient-held diaries used for monitoring values such as home based blood pressure monitoring booklets or patient instructional booklets for example for diabetic foot care. • Studies that focus on record use for specific and isolated transitions of care, such as hospital shift-change or discharge. Interventions such as a discharge summary or referral letter alone. • Studies that focus on single function of patient-held medical records such as medication prescriptions, lab results, blood pressure or blood sugar monitoring or out-patient registration papers.
Outcomes	• Outcome relevant to the quality of verbal and/or documented patient-specific information and	

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	communication. Standards of information and recording (completeness, accuracy and clarity) of the documented handover information was included. • Patients' and HCPs' views on how PHRs enabled/did not enable communication and documented information exchange was also included. Patients carrying records to consultation, availability of documented medical information for HCPs during consultation was also included. • Patient and HCP satisfaction with PHR, patient centred communication outcomes such as patient satisfaction, recall, understanding and adherence have been included. • Clinical outcomes and adverse events associated with quality of handover communication were also of interest, including (but not limited to) readmissions, diagnostic delays, healthcare utilisation or improved appointment rates, and death.	• Descriptive studies which describe the distribution of PHRs without information on outcomes of PHRs.

Legend: PHR=patient-held health record, LMIC=low and middle income countries, HIC=high income countries, HCP= health care HCP NCD=non-communicable diseases

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Search strategy

The search strategy (Supplementary file 1, S1) was developed for the electronic databases MEDLINE (Ovid) to identify journal articles. The search strategy was made following previous reviews regarding PHRs[26 28]. EMBASE (Ovid EMBASE Classic + EMBASE (1947 to present)) and CINAHL (via EBSCO (1980 to present) were searched for studies reporting PHR in LMICs. Additionally, citations were hand searched using relevant systematic reviews[19 28] to locate further relevant studies. Key search terms related to PHRs and LMICs were used to build the search strategy. The search terms used for PHRs were "patient-held book* or handbook* or card", "patient passport", "log book" and "home based records". Searches in the electronic databases were conducted from database inception until September 2020. No limits on language or publication year were applied during the search. Grey literature search was done by contacting authors of papers which reported the use of patient-held records, discussion with experts and using the website of World Health Organisation (WHO) and google scholar.

Study selection

The search results were uploaded to the reference management software Zotero to remove duplicates. The remaining studies' titles were independently screened by two reviewers (LJ and DB). The initial title screening was broad and retained all studies which referred to PHRs. Two reviewers (LJ and DB) independently screened the abstracts and full texts against the inclusion criteria. Any uncertainty surrounding the inclusion of a study or disagreement following discussions between reviewers were resolved through the assessment of an additional reviewer (SMH).

Data extraction

LJ reviewed all the data extraction tables. If there were any discrepancies during this sampling check, discussion took place and for final clarification a third impartial reviewer (JP or SMH) was consulted.

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Data were extracted from published studies using a data extraction form in Microsoft Excel and included details of authors, study settings and country, the objectives and features of the PHR, use of PHRs, and outcomes.

Quality assessment

Quality assessment was done by two reviewers (LJ and DB) using the Mixed Methods Appraisal Tool (MMAT).[34] This provided an overall description of the quality of studies and used descriptors 'low', 'medium' and 'high' to provide an indication of the quality of the included studies. For quantitative, qualitative and cross-sectional studies, all five criteria needed to be met to be classified as 'high quality'. Studies that met three to four criteria were classified as 'medium quality' and studies that met one or two criteria were regarded as 'low quality'. For mixed-methods studies, the overall score was dependent on the lowest score of each of the study components (qualitative and quantitative); therefore, scores were determined by the quality of the weakest component. In case of disagreements on quality assessment, the final decision was taken after discussion with an independent third reviewer (JP or SMH).

Data analysis

Given the diversity of population, settings, the non-uniformity of intervention, and the variations in outcome measures, a statistical meta-analysis was not appropriate. Hence, the review used a narrative synthesis. Papers in the review have been summarised descriptively as follows (Table 2).

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Table 2: Description of results

Result	Description of the result
Patients carrying the PHRs to HCP visits	Presented as frequencies or number of patients carrying the records to visits or as prevalence of written clinical information availability for HCPs at visits.
HCPs' recording of information and Quality of information recorded	HCPs' recording of information in the PHR is presented as information availability for patients on leaving the facility or patient/HCP self-reported availability of documented information. The quality of information recorded for this review is defined as the completeness of the information on key elements such as diagnosis, medication/treatment details and follow-up information, clarity or legibility of the information recorded and accuracy of the information being recorded based on treatment guidelines (For example; mismatch of diagnosis and treatment prescribed)/comparison of data with facility based records. Completeness data are reported as frequencies and stand-alone or comparative data based on key missing components.
Utility of PHR to HCPs and patients	For the review, utility as perceptions of patients and HCPs about using PHRs, satisfaction with use, usability in terms of ease of reading the records, and the functions they serve (for information exchange, clinical information recording at each HCP visit, patient education or as an aide memoire for patients).

Legend: PHR=patient-held health record, HCP=health care HCP

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Patient and public involvement

This research was done without patient or public involvement.

Ethics

Ethics approval was not applicable

RESULTS

Study selection

The findings of the search strategy are summarized in Fig 1 as a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart.[35] 2286 studies were screened and six studies were included in the final review. The reasons for exclusion of studies are presented in Fig 1.

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Characteristics of included studies

There were studies from Zambia (n=x)[36], Lesotho (n=x)[37], South Africa (n=x)[38 39], China (n=x)[40], and Mongolia (n=x)[41] (Table 3). The designs of included studies were mixed methods (n=1)[41], observational (n=2)[36 37] and qualitative (n=3)[38-40]. Three studies[37 40 41]evaluated PHRs that were implemented and routinely used by the health department in those countries and one study evaluated PHR using diabetes mellitus and hypertension as tracer conditions for NCDs. All the studies described the use of PHRs in outpatient settings, while none measured changes in health outcomes following the use of PHRs.

Quality appraisal

Of the six studies included, five were judged to be of 'medium quality' and one was of 'low quality' (Supplementary file1, S2). For the three qualitative studies[38-40], the methods and data collection were appropriate to answer the research question. However, the studies did not mention the use of a specific method of data analysis e or provide a rationale for using an analytic method. Therefore, coherence between data sources, collection, analysis and interpretation could not be fully assessed. One mixed methods study[41] did not adequately explain the method of qualitative data analysis and was marked down on that criterion. Of the two cross-sectional studies, one[36] provided no information on the sampling strategy and hence it was not possible to make judgements on the representativeness of the sample. Further, no information was reported regarding the non-response rate. The other cross-sectional study provided information on sampling strategy, a reasonable description of the target population and response rate. However, the study was marked down due to insufficient information on measurement variables.[37]

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Table 3: Characteristics of included studies

Author, year, LMIC	Study design, sample size	Study objective	Settings	Participants	PHR description	Results
Van der Hoek [36], 1995, Zambia	Cross sectional (n=176,n=184)	To describe the views of patients on the newly introduced health passports by health management in Sesheke district, Zambia	Household surveys (n=2), conducted in the catchment areas of one hospital and two health centres.	Patients	Hardcover booklet with 32 pages known as a health passport, size-14.5*10cm, Demographic information on the cover page. Referral letters and discharge letters are replaced with health passports.	68% of the respondents bought a health passport within 4 months of their introduction and 91% within 16 months. 36% (64/176) and 30% (55/184) of the respondents felt that the passports were expensive.
Henbest[37], 1995, Lesotho	Cross sectional (n=878) Patients(n=691) Nurse clinicians (n=74) and doctors(n=104)	To inform the recommendations for the use of PHR in healthcare services in changing Africa, using the PHR in Lesotho.	Healthcare Centres and household survey	Patients, nurses and doctors	PHR, in the form of a health booklet about the size of a passport (10.5*15cm) with a brightly coloured plasticised cardboard cover. The front cover contains the patient's name, address, date of birth and instruction on the record's use. The back cover has the common symptoms and signs of tuberculosis. The inside printed pages contain forms for recording information such as immunisations, previous significant history drug sensitivity and screening activities. There are 16 blank pages for HCP notes. When	89% of patients preferred to have a PHR than a facility based record. 79% of doctors and 70% of nurses rated the use of PHR as excellent. Doctors and nurses perceived that having a PHR reduces unnecessary repetitions of tests and treatments. They also felt that availability of patient medical information from other HCPs leads to less errors.

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					one record is full, it is stapled to the next record.	
Norden[38],2004 South Africa	Qualitative study Focus group discussions with patients (n=24)	To understand the patients' thoughts on the PHR introduced. It was introduced due to the difficulty in retrieval of medical information at HCP visits from facility based records.	Primary care clinic	Patients	Pocket size booklet with 10 pages (14*8 cm) Demographic information, problem list and allergies recorded in the first page. Blood pressure, blood glucose, body mass index, peak expiratory flow rates are recorded in the subsequent visits. Prescribed treatment and follow up dates are also documented.	Patients felt that the records function as 'medical identification' and are useful in an emergency, provide important background information, and that they enable continuation of care at other facilities. Patients reported that PHRs could motivate them to act on the advice given, and the records also served as a reminder to take their medication and about the dates of follow-up visits.
Kerry[39] , 2005, South Africa	Quality assurance methods with Focus group discussions (n=13) number of participants not stated	To assess, document and improve the PHR System in the Emtshezi Subdistrict, South Africa.	Health care facilities at sub-district level	Patients, nurses, doctors, administrators and pharmacists	Proposed better PHR: A single booklet PHR for every patient in the district. Size – 10 x 21cm named as Health Book; at least 32 pages for clinical notes; a problem list; pages for laboratory and X-ray results and incorporation of the Health Card for Women, TB Card and Antenatal Card. The booklet should have a plastic cover.	Patients and HCPs reported that communication of clinical information between health facilities was poor. HCPs described there were difficulties in retrieval of information and recording at hospitals, the current PHRs were unstructured and multiple PHRs were brought by patients. Patients and HCPs felt that having a single, common PHR to be the definitive ambulatory health record for every patient at district level. Feedback from the FGDs were used to design a better PHR.

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Chen[40] et al, 2014,China	Qualitative study (semi-structured interviews and observations Doctors (n=4), patients(n=13), family members (n=4)	To examine and explore practices surrounding PHR in a setting where medical records have long been managed by patients: the Chinese healthcare system.	Out-patient departments of hospital.	Patients, family members and doctors	Pocket size book which had chief complaint and history handwritten by the clinician. Diagnosis and prescription are documented on the electronic health system and printouts of the same are provided for the patients.	The findings suggest that through engaging in practices of managing and sharing records, patients were able to obtain familiarity with their own records and to provide necessary assistance to locate information for HCPs to use at the point of care. Clinicians felt that a verbal history by the patient was insufficient, as was a history based solely on their own clinic's records.
Ibrahim[41] et al, 2019, Mongolia	Mixed methods, (n=395)	To describeMongolia's universal patient-held health booklets and its use, explore patients' views on using them and explore training and protocols using tracer conditions diabetes and hypertension.	Out-patient departments of two public hospitals	Patients,	Booklet with 40 pages (14.8*21cm). Demographic information on the cover page, history or handover information in subsequent pages.	94% of patients had PHRs with them at OPD visits. Provision of written information for the three categories of key information items (i.e. diagnosis, prescription/management and follow-up information) for the consultation was low at around 37% in the PHR;- 61% had notes related to medication required and 40% contained information about follow-up. Patients (316, 80.0%) reported consulting their PHRs and associated 'documents' at home. They reported using them for understanding their own condition and communicating it to others

Legends: PHR=patient-held health record, HCP=healthcare HCP, FGD=focus group discussion

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Synthesis of results from included studies

Overall, the included studies were inadequate to demonstrate a clear benefit of using PHRs to improve information availability to HCPs and patients. However, based on the patients' perspective, PHRs were beneficial to the management of their own care. Further, based on the HCPs' perspective PHBs improved the availability of key clinical information for providing care.

Patients carrying their PHRs

Of the six studies, two reported data on the prevalence of patients bringing their PHR to subsequent visits. In Mongolia, Ibrahim et al[41] reported that 94% (373/395) of chronic NCD patients brought their PHB to their outpatient visits. This was measured by examining the documents brought to the visit by the patients.

In Lesotho, Henbest et al[37] used a survey with patients and HCPs to report on the practice of carrying records to consultations and HCPs' opinions on the availability of PHRs at consultations. Patients' self-reported practice of carrying PHRs to healthcare consultations was 71% (n=672) and HCPs self-reported the availability of the PHR at visits as 62% (doctors, n=81) and 59% (nurse clinicians, n=68) respectively.

In China, Chen et al [40] used interviews with, and observation of, patients to report patients' perspectives on, and patterns of, carrying records to consultations. They found that the patients with chronic NCDs had different PHRs, according to particular conditions/diseases, and clinics or hospitals. Patients often carried only specific PHRs to consultations based on the condition/HCP/clinic they visit.

HCPs' recording information and the Quality of information recorded

Of the six studies, only one (from Mongolia[41]) evaluated HCPs' documenting of information in PHRs and the quality of the information recorded for patients with diabetes and hypertension. The overall written information, across three categories (diagnosis, prescription and follow-up), within the PHR

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after the consultation was 37% (n=367). The completeness of PHRs with respect to medication and follow up information were 61% and 40%, respectively. No studies reported the clarity, legibility or accuracy of the recorded information.

Utility of PHR to HCPs

In Mongolia, Ibrahim et al[41] did not formally assess HCPs' perspectives on the use of PHR, but they did find that HCPs in the two public outpatient hospitals wrote in the booklets in spite of also having to enter information in other electronic record systems. Further, more than three-quarters (77.4%, n=106) of patients reported that private HCPs they had visited recorded clinic notes in the PHR.

In Lesotho, Henbest et al[37] described HCPs' satisfaction with using PHRs. This was measured in terms of preference for PHRs, practicality (size, durability and confidentiality) and comparison with facility based records (quality of care, access to patient information). Both doctors and nurse clinicians reported a preference for PHRs over facility-based paper records. The HCPs perceived that having a PHR contributed to improving quality of care by preventing the unnecessary repetition of tests and treatments (86% of nurse clinicians (n=71) and 88% of doctors (n=89)) and that the availability of medical information from other HCP visits helped in the clinical management of patients (78% of nurse clinicians (n=71) and 84% of doctors (n=89)). The qualitative study by Chen et al [40] from China[40], reported that a patient's verbal history of past medical information was perceived as insufficient for providing effective care by doctors. Further, access to previous documented clinical information was perceived by doctors as essential to make appropriate clinical decisions. The other two studies[38 39] did not report the HCPs' perspectives on PHRs.

Protocols and HCP Training

The study from Mongolia[41] explored the role of protocols and policy directions, with a Government order reinforcing the use of PHRs. However, no written guidelines or training materials regarding the use of PHRs or training for clinical handover were reported by HCPs. Another study reported one day training for health workers, with practical instructions on how to use the PHR.[36]

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Utility of PHRs for patients

All included studies reported some use of the information in the PHRs by patients. Ibrahim et al[41] described how Mongolian patients (316, 80.0%) consulted their PHRs at home. Patients reported that written information from doctors was important for their own understanding of their condition. Van der Hoek et al[36] reported that some Zambian patients (n=176, 16% (at 6 months) and n=183, 14% (at 18 months)) preferred to keep their records at the clinic itself, due to a fear of damaging the records at home but they did not report whether the patients used the PHRs at home or not. Henbest et al[37] found that most (n=691, 89%) of Lesotho (South African) patients preferred to have a PHR and perceived the record as belonging to them. They reported that having a PHR was important to making information available to the other HCPs they visited. They highlighted that this was very useful in case of travel and emergencies. Norden et al [38] also reported similar findings for South African patients. Patients viewed their PHR as a document containing their own medical information and a tool for continuation of care between HCPs. Additionally, the PHR helped these patients and their families to remember how to manage their own care at home. In their qualitative study, Chen et al [40] found that Chinese patients regarded PHRs as useful. They reported that patients read their documents, organised them chronologically and brought them to each subsequent visit.

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DISCUSSION

Summary of findings

This systematic review identifies that there is limited literature from LMICs on the use of PHRs for improving information availability to HCPs and patients for handover communication. This paucity of published research limits our ability to draw conclusions on the benefits or challenges of PHRs in care for patients with NCDs. Importantly, none of the studies included in the review investigated, reported or evaluated changes in health outcomes after using PHRs in LMICs. Consistent with studies of maternal and child PHRs[42-44], in general, patients brought their PHRs to each clinical consultation.

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The use of PHRs in the form of a booklet in Mongolia and China and attaching medical reports (in the form of papers) to the booklet reduced the risk of losing them. Although only one study addressed the inadequate completeness of the information documented by HCPs, and a lack of protocols and training on the use of PHRs, are likely to be prominent issues in all the settings as evidenced by the wider literature on training and protocols for medical records documentation.[45-47] The PHRs were perceived as useful by HCPs for including documentation of the medical history of patients, thus improving availability of medical information. From the patient perspective, the PHR functioned as a reminder for medication, improved self-care efficacy and enabled continuity of care by providing relevant clinical information to all HCPs in the patient care. Consistent findings on the use of PHR in improving self-care for management of chronic conditions were reported and equally reflected in HIC studies.[27 28]

Implications for research and practice

The findings from the review suggest that PHRs may be particularly useful in health systems where little or no medical record keeping occurs at outpatient level, where record retrieval is difficult, or where multiple HCPs may be involved in a person's care for chronic conditions. This is especially important in LMICs as most have pluralistic health systems and no 'gate-keeping', so that patients often change their HCPs, or shop around, and receive care at both public and private facilities.[15 16 45] Patients' provision of a verbal medical history as the only means of transferring previous clinical information during HCP visits is, however, insufficient. In LMICs patients' communication of past medical history may be adversely affected by a lack of health literacy, inability to articulate the clinical procedures they received, and a lack of confidence against a background of profound power differentials between patients and HCPs.[48-50]

A well completed PHR may in principle improve medical information availability for HCPs. However, no trials which assessed the availability of medical information for HCPs were reported from LMICs. In Germany, Straßner et al[51] conducted a trial investigating the utility of PHRs for asylum seekers in

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reception centres for improving medical information availability for HCPs. The trial found that, a PHR for asylum seekers increased the availability of health information for HCPs (aOR 4.22, 95% CI: 2.64 to 6.73), reduced missing information (aOR 0.89, 95% CI: 0.42 to 1.88) and reduced HCP dissatisfaction (aOR 0.43, 95% CI: 0.16 to 1.14) .[51]

Patients, especially those with chronic diseases, usually carry records they have to HCP visits, especially if they know HCPs will use them. [15 16 45 52] However, in the absence of an organised PHR, patients either bring very little information such as a prescription alone (common in LMICs), or they bring all their previous pieces of paper that form their home held medical records, making it difficult for HCPs to find the relevant information.[16] Many chronic disease patients have multi-morbidities and often have numerous papers or clinic specific PHRs from the different clinics/hospitals they visit, but they often only carry notes from previous visits to the clinic they are going to visit.[28] Thus, the HCPs do not receive the information about medication and management plans produced by other doctors in other clinics or hospitals. Therefore, it is necessary to develop generic/universal and pragmatic PHRs and create awareness of the importance of using a single generic PHR for all healthcare visits, irrespective of the clinic/hospital, so that all HCPs have the necessary medical information available to make the best clinical decisions.

The WHO recommends that ideally a PHR should be used as complementary to a facility based medical record.[53] Prior reviews from HIC, showed that when HCPs had to enter information in facility based records (electronic or paper based) and PHRs, this multiple recording was an unwelcome burden.[54 55] However, documented discharge information for patients /families (e.g., summaries/letters) from HCPs has been found to reduce post-discharge complications[56] and re-admissions,[57] and to improve patient satisfaction[58] and health outcomes.[59] Arguably, it is more efficient and accurate if there would be one record system accessible and useful to both patients and HCPs.

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Many LMICs are progressing with the implementation of electronic health records which may enable health systems to overcome difficulties of retrieving information from paper-based facility-based medical records.[60 61] Despite such advancements in technology and information technology capability in some middle-income countries, paper records will remain the prominent form of medical records and are unlikely to be completely replaced by electronic records at least for the near future. Furthermore, the issues of integrating electronic health records across different levels of care in health systems and public-private information exchange are often not addressed in existing health information management systems.[62]

Strengths and limitations

Our systematic review has several strengths and limitations. The review protocol was registered at PROSPERO and we followed PRISMA guidelines in our systematic review. A comprehensive search strategy was employed to locate studies from LMICs. Furthermore, our search PHRs from the grey literature and discussion with experts in the field helped us to identify generic PHRs from Malawi,[63] Namibia[64] and South Africa.[65]However, these PHRs have not been evaluated systematically, and we have limited information on the usefulness of these records. However, our study also has some limitations. Given the small number of relevant articles, we found very few RCTs to base conclusions of robust experimental findings. Scarce data on generic PHRs from LMICs limits our ability to interpret the usefulness of these records in improving health outcomes of patients with NCDs.

CONCLUSION

Available data on the use of non-maternity PHR for handover communication between HCPs and HCPs with patients in LMICs are sparse. Existing studies on PHR in LMICs are of medium to low quality, and heterogeneous in terms of study design, population and context. Further, health outcome assessments after the introduction of PHRs are not available. The available studies show that patients value PHRs for their own use and are likely to carry them to HCPs. HCPs, in turn, consider that PHRs enhance documented information transfer from previous visits specifically when

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there is difficulty in retrieving past medical information. With LMICs progressing with electronic health record implementation, the role of paper based PHRs in overcoming issues of integration and for improving adherence to effective self-care needs re-evaluating. Future research should explore minimum information requirements for major conditions, the format and types of PHRs of most utility to patients and HCPs (electronic or paper-based), and barriers and facilitators to effective use of PHRs across all HCPs for any one patient. Further, the content and HCP training and protocols need to be developed and evaluated for the effective scale-up of interventions to promote the effective use of PHRs for continuity of care and patient self-care.

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Declaration of interest

We declare no competing interest.

Author contributions

JLM, PJ, SG and SMH conceived the article. JLM performed the literature search. JLM, BD, JP collected data from individual studies and interpreted the data. JLM wrote the first draft of the paper. SMH, SG, PJ, AL and CH revised the first and subsequent drafts. All authors contributed to interpretation of the findings and revised the manuscript for important intellectual content.

Data sharing

No additional data available

Fig 1. Legends

LMIC=Low and middle income countries, CVD=Cardiovascular disease, PHR=Patient-held health record, NCD=Non-communicable disease

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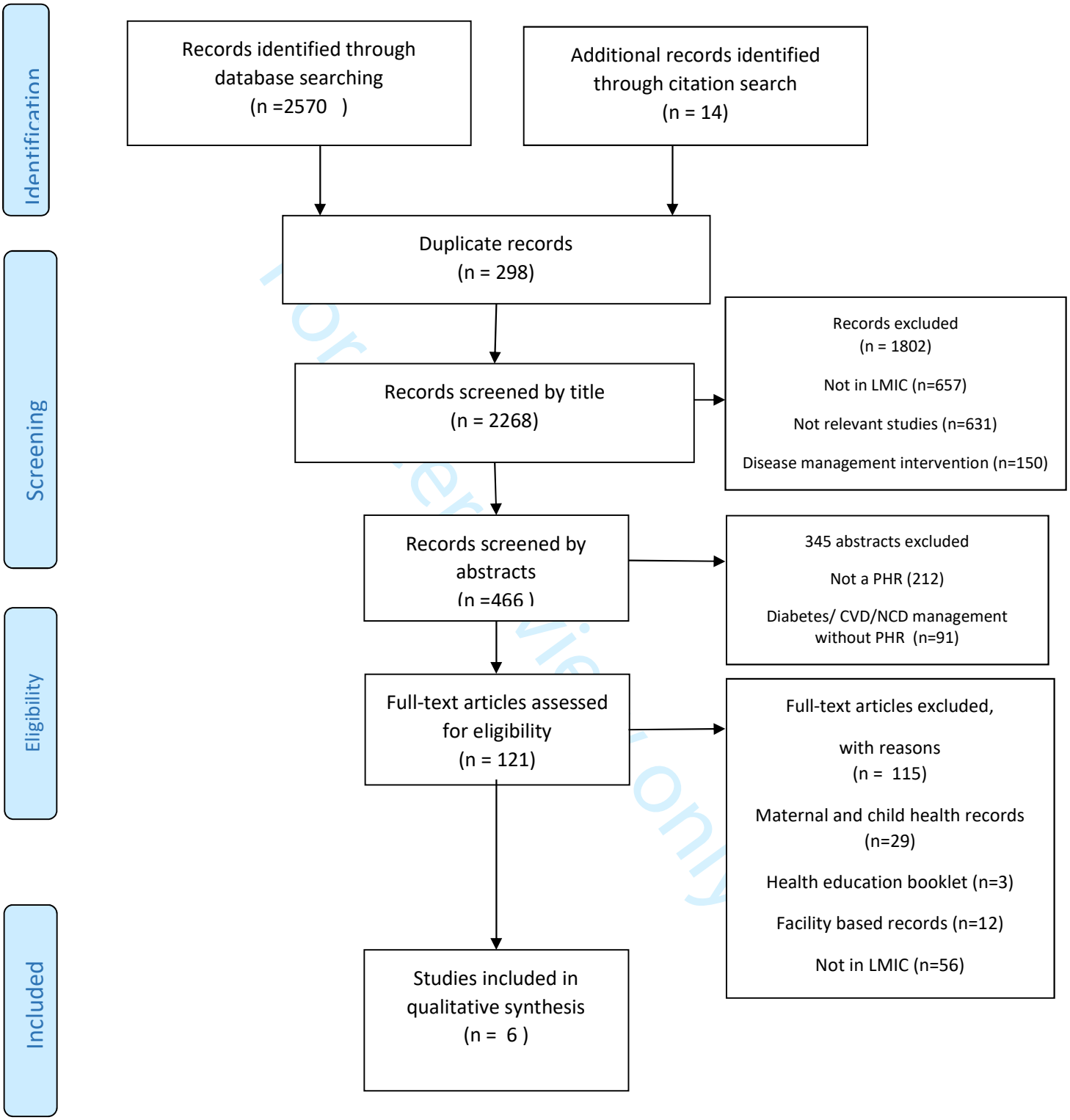


Fig 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram

SUPPLEMENTAL MATERIAL

S1.Search strategy in EMBASE

Patient-held records

1. (exp medical records/or exp medical records, problem-oriented/or exp medical records systems, computerized/or exp nursing records/or (((medical or health) and record*).mp. or (case note* or case record*).mp.) or (record* or book* or handbook* or card*).mp AND ((held adj3 (patient or parent or person or woman or man or family or consumer)).mp. or (carried adj3 (patient or parent or person or woman or man or family or client or consumer)).mp.)) OR ((log-book* or logbook*).mp. or home based record*.mp. or passport*.mp. or personal health record*.mp.)
2. ((patient-held or patient-held or home based or home-based or personal child or "mother and child" or "maternal and child" or mother-child) adj2 (record* or book* or handbook* or card*)).mp
3. ((patient access or personal or patient-held) adj2(record*or portal * or health record* or information* or prescription* or electronic record*)).mp
4. 1 or 2 or 3

Low-and Middle-income countries

5. exp developing country/ or exp medically underserved/ or developing countr$.mp. or medically underserved area$1.mp. or low income countr$.mp. or middle income country.mp. or low resource.mp. or resource poor.mp. or global.mp. or exp Africa/ or exp "South and Central America"/or exp asia/ or exp Caribbean islands/ or exp pacific islands/ or exp eastern Europe/ or exp Indian Ocean/or south america$1.mp. or Africa$1.mp. or Caribbean.mp. or central America$1.mp. or south America$1.mp. or eastern Europe$1.mp. or pacific island$.mp. or Indian ocean island$.mp. or asia.mp. or Afghan$.mp. or Bangladesh$1.mp. or Benin$.mp. or Burkina Faso.mp. or Burkinabe.mp. or Burundi$.mp. or Cambodia$1.mp. or Central African.mp. or Chad$.mp. or Comor$.mp. or Congo$.mp. or Eritrea$1.mp. or Ethiopia$1.mp. or Gambia$1.mp. or Guinea$1.mp. or Haiti$.mp. or Kenya$1.mp. or Korea$1.mp. or exp North Korea/ or Kyrgyz$.mp. or Liberia$1.mp. or Madagascar.mp. or Malagasy.mp. or Malawi$.mp. or mali$.mp. or mozambi$.mp. or Myanmar$.mp. or Nepal$.mp. or Niger$.mp. or Rwanda$1.mp. or Sierra Leone$.mp. or Somalia$1.mp. or Tajik$.mp. or Tanzania$1.mp. or Togo$.mp. or Uganda$1.mp. or Zimbabwe$.mp. or Angola$1.mp. or Armenia$1.mp. or Beliz$.mp. or Bhutan$.mp. or Bolivia$1.mp. or Cameroon$.mp. or Cape Verde$.mp. or Congo$.mp. or "Côte d'Ivoire".mp. or Ivory Coast.mp. or Ivorian.mp. or Djibouti.mp. or Egypt$.mp. or El Salvador.mp. or Salvadoran.mp. or Fiji$.mp. or Georgia$1.mp. or Ghana$.mp. or Guatemala$1.mp. or Guyan$.mp. or Hondura$.mp. or Indonesia$1.mp. or India$1.mp. or Iraq$1.mp. or Kiribati.mp. or Kosov$.mp. or Lao$.mp. or Lesotho.mp. or Marshall Islands.mp. or Marshallese.mp. or Mauritania$1.mp. or Micronesia$1.mp. or Moldov$.mp. or Mongolia$1.mp. or Morocc$.mp. or Nicaragua$1.mp. or Nigeria$1.mp. or Pakistan$1.mp. or Papua New Guinea$1.mp. or Paraguay$.mp. or Philippines.mp. or Filipino.mp. or Samoa$1.mp. or sao tome$.mp. or Senegal$.mp. or Solomon Island$.mp.

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1. 4 and 5

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S2. Table summaries of MMAT quality appraisal assessments for included studies

Author, year, LMIC	Type of study	4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)
Van der Hoek [23], 1995, Zambia	Quantitative descriptive	Can't tell	Can't tell	Yes	No	Yes	2
Henbest[24], 1995, Lesotho	Quantitative descriptive	Yes	Yes	Can't tell	Yes	Yes	4
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
Norden[25],2004 South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Kerry[26] , 2005, South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Chen[27] et al, 2014,China	Qualitative	Yes	Yes	Yes	Yes	Can't tell	4
		4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)

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Ibrahim[28] et al, 2019, Mongolia	Quantitative descriptive	Yes	Yes	Yes	Yes	Yes	5 (Has description of non-responders and confounders)
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
		5.1. Is there an adequate rationale for using a mixed methods design to address the research question?	5.2. Are the different components of the study effectively integrated to answer the research question?	5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?	
	Mixed methods	Yes	Can't tell	Yes	Can't tell	Yes	3(Integration may have occurred in the interpretation of results-It is unclear, cannot tell if there is divergent findings)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1, Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5,7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7,8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file, S1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10,11

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12,13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13 and Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 and Supplementary file, S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	18-20
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Na
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20

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Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	23
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file, S1
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A systematic review on the use of patient-held health records in low and middle-income countries.

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Manuscript words **3938 (excluding abstract, tables, figures, references)**

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ABSTRACT

Objective

To review the available evidence on the benefit of patient-held records(PHRs), other than maternal and child health records, for improving the availability of medical information for handover communication between healthcare providers (HCPs) and/or between HCPs and patients in low and middle income countries (LMIC).

Methods

The literature searches were conducted in PubMed, EMBASE, CINAHL databases for manuscripts without any restrictions on dates/language. Additionally, articles were located through citation checking using previous systematic reviews and a grey literature search by contacting experts, searching of the World Health Organisation (WHO)website and Google Scholar.

Results

Six observational studies in four LMICs met the inclusion criteria. However, no studies reported on health outcomes after using PHRs. Studies in the review reported patients' experience of carrying the records to HCPs (n=3), quality of information available to HCPs (n=1), and the utility of these records to patients (n=6) and HCPs (n=4). Most patients carry PHRs to healthcare visits. One study assessed the completeness of clinical handover information and found that only 41% (161/395) of PHRs were complete with respect to key information on diagnosis, treatment and follow-up. No protocols or guidelines for HCPs were reported for use of PHRs. The HCPs perceived the use of PHRs improved medical information availability from other HCPs. From the patient perspective, PHRs functioned as documented source of information about their own condition.

Conclusion: Limited data on existing PHRs make their benefits for improving health outcomes in LMICs uncertain. This knowledge gap calls for research on understanding the dynamics and outcomes of PHR use by patients and HCPs and in health systems interventions.

Keywords -**International health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH**

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Study design that enabled the accumulation and presentation of current evidence on the use of patient-held health records other than maternal and child health records in low and middle income countries.
- Thorough literature search of three major electronic databases and reporting as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.
- Inconclusive results due to no evidence from trials.
- Small number of studies and no reporting on health outcomes, making it difficult to determine usefulness for patients with non-communicable diseases.

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INTRODUCTION

To meet sustainable development goals[1] and deliver quality care[2], health systems in low and middle income countries (LMICs) need to be redesigned and strengthened.[3] In line with this agenda, the World Health Organization (WHO) has formulated a health systems framework that describes six building blocks for health systems; of these, good service delivery comprises the delivery of effective, safe, quality health care and continuity of care is an important characteristic for delivering quality care.[4]

In many LMICs, healthcare delivery is usually organised around acute and single episode care, from different facilities and health care providers (HCP).[5] Such a model of healthcare delivery is inadequate due to increasing non-communicable disease (NCD) burden in LMICs.[6] Continuity of care needed for efficient and effective care, is simply defined as "the seamless provision of healthcare between settings and over time".[7] Informational continuity represents the cornerstone of continuity of care and is defined as the use of information on past events and personal circumstances to make current care appropriate for each individual.[8] Handover communication not only relates to the process of passing patient-specific health information between HCPs, from one visit to another, but also from HCPs to the patient and family for ensuring patient care-continuity and self-management.[9]

Complete and accurate clinical information about previous management and treatment is necessary for managing all patients but especially patients with chronic conditions due to a need for on-going care from a range of HCPs.[10] Inadequate documented handover for HCPs often results in adverse events[11 12] or increased in-patient visits[13], repeated tests or examinations leading to costly healthcare, burdensome to patients and a waste of resources.[13] The ability to view patients' medical records, across healthcare visits and facilities helps to facilitate health information exchange.[14] The absence of out-patient medical records,[15 16]difficulties in retrieval of facility based records for out-

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patient visits,[17] and lack of integrated patient electronic medical records between departments and between health facilities in LMICs contribute to poor informational continuity for HCPs.[18]

A patient-held health record (PHR) can be a viable solution for improving sharing of documented medical information across HCPs.[19] For the purpose of this review, PHRs are formal records given to patients in the form of booklets, which contain patients' medical information. Our hypothesis is that such PHRs with documented medical notes from HCPs can be used as a tool for improving medical information availability for HCPs.[15 20]

Paper-based PHRs are used extensively in maternity care, as home-based records or maternal and child health handbooks for sharing information across antenatal, labour and post-natal care by all HCPs even in LMIC settings.[21] Three systematic reviews on maternity and child records indicate that PHRs improve HCP-patient communication and facilitate referrals, and both HCPs and women give positive feedback on their use.[22-25]

There are several systematic reviews[19 26-29] of PHRs for chronic conditions from high income countries (HIC), which evaluated their usefulness in cancer care,[28] and chronic conditions,[26 27] and for people with severe mental illness[29] or undocumented immigrants.[19] These condition-specific PHRs were designed to mainly improve communication between HCP and patients.[30] The reviews found that PHRs may improve a patient's sense of control and empowerment, leading to better involvement in their own care.[28 29]

There is no systematic review of PHRs for conditions other than maternal and child health from LMICs, while there is evidence that generic PHRs do exist in some LMICs.[31] A study in India found that patients were carrying unstructured sheets of paper to out-patient visits and patient-held health booklets were an acceptable and pragmatic intervention to improve information exchange for chronic NCD patients by patients and HCPs.[15]

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The current review aimed to investigate the benefits of generic PHRs used in LMICs to ensure the availability of medical information for clinical handover communication and continuity of care between HCPs or outpatient health facility visits, with or without their use for communication between HCPs and patients. This review investigated non-maternal-child health PHRs, designed to record patient histories and healthcare information across a range of conditions and HCPs, including primary and secondary care and not specific to one facility or one speciality.

METHODS

The protocol for this review is registered with the PROSPERO International Prospective Register of Systematic Reviews (CRD42019139365)[32].

Selection criteria

The published literature was searched for quantitative randomised controlled trials (RCTs), cluster RCTs, quasi-experimental studies, cross sectional and pre-post study designs, qualitative and mixed-method studies. Case reports, commentaries and review articles were excluded studies. The eligibility criteria are provided in Table 1.

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160

161 Table 1 Inclusion and exclusion criteria

Title	Inclusion criteria	Exclusion criteria
Population	• Participants (patients, carers and/or health care HCPs) from any health background, of 18 years or more, using a PHR were considered. • All healthcare settings within LMICs (as defined by the World Bank 2018)[33] were considered.	• Children and pregnant women • Patients/ carers/HCPs from HIC
Intervention	• Studies which include design, implementation or evaluation of formalised PHRs , with the purpose of improving information exchange and communication between visits to the same facility for HCPs, across different health care facilities (primary to secondary/vice versa), and documented information from HCPs to patients (for patients' own care). • Studies which describe PHRs which are clinically focused and person specific or generic PHRs. Studies which describe condition specific PHRs, they should include NCDs such as Diabetes Mellitus, hypertension, cardiovascular diseases, chronic respiratory diseases or cancer. • PHRs are defined as any formal medical document in the form of a booklet held by patients, which can be used across health care settings or visits, contain patient histories and healthcare information to guide healthcare workers providing care.	• Facility based medical records, which are for HCP use. • Home based records used for maternal and child health care. • Patient-held diaries used for monitoring values such as home based blood pressure monitoring booklets or patient instructional booklets for example for diabetic foot care, or patient-held records for single communicable disease such as tuberculosis. • Studies that focus on record use for specific and isolated transitions of care, such as hospital shift-change or discharge. Interventions such as a discharge summary or referral letter alone. • Studies that focus on a single function of patient-held medical records such as medication prescriptions, lab results, blood pressure or blood sugar monitoring or out-patient registration papers.
Outcomes	• Outcome relevant to the quality of verbal and/or documented patient-specific information and	

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	communication. Standards of information and recording (completeness, accuracy and clarity) of the documented handover information was included. • Patients' and HCPs' views on how PHRs enabled/did not enable communication and documented information exchange was also included. Patients carrying records to consultation, availability of documented medical information for HCPs during consultation was also included. • Patient and HCP satisfaction with PHR, patient centred communication outcomes such as patient satisfaction, recall, understanding and adherence have been included. • Clinical outcomes and adverse events associated with quality of handover communication were also of interest, including (but not limited to) readmissions, diagnostic delays, healthcare utilisation or improved appointment rates, and death. Intermediate outcomes such as blood pressure/blood glucose monitoring, medication management/reconciliation and/monitoring of lab values were included.	• Descriptive studies which describe the distribution of PHRs without information on outcomes of PHRs.

Legend: PHR=patient-held health record, LMIC=low and middle income countries, HIC=high income countries, HCP= health care HCP NCD=non-communicable diseases

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Search strategy

The search strategy (Supplementary file 1, S1) was developed for the electronic databases MEDLINE (Ovid) to identify journal articles. The search strategy was made following previous reviews regarding PHRs[26 28]. EMBASE (Ovid EMBASE Classic + EMBASE (1947 to present)) and CINAHL (via EBSCO (1980 to present) were searched for studies reporting PHR in LMICs. Additionally, citations were hand searched using relevant systematic reviews[19 28] to locate further relevant studies. Key search terms related to PHRs and LMICs were used to build the search strategy. The search terms used for PHRs were "patient-held book* or handbook* or card", "patient passport", "log book" and "home based records". Searches in the electronic databases were conducted from database inception until September 2020. No limits on language or publication year were applied during the search. Grey literature search was done by contacting authors of papers, which reported the use of patient-held records, discussion with experts and using the website of World Health Organisation (WHO) and google scholar.

Study selection

The search results were uploaded to the reference management software Zotero to remove duplicates. Two reviewers (LJ and DB) independently screened the remaining studies' titles. The initial title screening was broad and retained all studies that referred to PHRs. Two reviewers (LJ and DB) independently screened the abstracts and full texts against the inclusion criteria. Any uncertainty surrounding the inclusion of a study or disagreement following discussions between reviewers were resolved through the assessment of an additional reviewer (SMH).

Data extraction

Two reviewers (DB and PJ) independently extracted data from the included studies. LJ reviewed all the data extraction tables. If there were any discrepancies during this sampling check, discussion took place and for final clarification a third, impartial reviewer (JP or SMH) was consulted.

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Data were extracted from published studies using a data extraction form in Microsoft Excel and included details of authors, study settings and country, the objectives and features of the PHR, use of PHRs, and outcomes.

Quality assessment

Quality assessment was done by two reviewers (LJ and DB) using the Mixed Methods Appraisal Tool (MMAT).[34] This provided an overall description of the quality of studies and used descriptors 'low', 'medium' and 'high' to provide an indication of the quality of the included studies. For quantitative, qualitative and cross-sectional studies, all five criteria needed to be met to be classified as 'high quality'. Studies that met three to four criteria were classified as 'medium quality' and studies that met one or two criteria were regarded as 'low quality'. For mixed-methods studies, the overall score was dependent on the lowest score of each of the study components (qualitative and quantitative); therefore, scores were determined by the quality of the weakest component. In case of disagreements on quality assessment, the final decision was taken after discussion with an independent third reviewer (JP or SMH).

Data analysis

Given the diversity of populations, settings, the non-uniformity of interventions, and the variations in outcome measures, a statistical meta-analysis was not appropriate. Hence, the review used a narrative synthesis. Papers in the review have been summarised descriptively as follows (Table 2).

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206 Table 2: Description of results

Result	Description of the result
Patients carrying the PHRs to HCP visits	Presented as frequencies or number of patients carrying the records to visits or as prevalence of written clinical information availability for HCPs at visits.
HCPs' recording of information and Quality of information recorded	HCPs' recording of information in the PHR is presented as information availability for patients on leaving the facility or patient/HCP self-reported availability of documented information. The quality of information recorded for this review is defined as the completeness of the information on key elements such as diagnosis, medication/treatment details (including lab values) and follow-up information, clarity or legibility of the information recorded and accuracy of the information being recorded based on treatment guidelines (For example; mismatch of diagnosis and treatment prescribed)/comparison of data with facility based records. Completeness data are reported as frequencies and stand-alone or comparative data based on key missing components.
Utility of PHR to HCPs and patients	For the review, utility as perceptions of patients and HCPs about using PHRs, satisfaction with use, usability in terms of ease of reading the records, and the functions they serve (for information exchange, clinical information recording at each HCP visit, patient education or as an aide memoire for patients).

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207

208 Legend: PHR=patient-held health record, HCP=health care HCP

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Patient and public involvement

This research was done without patient or public involvement.

Ethics Approval

No ethical approval was needed because data from previous published studies is being used.

RESULTS

Study selection

The findings of the search strategy are summarized in Fig 1 as a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart.[35] 2286 titles were screened and 466 abstracts were included. 121 full papers were reviewed and six studies were included in the final review. We excluded 115 articles because: studies that described PHRs but were not conducted in LMICs (56); studies that described maternal or child health records in LMICs (29); papers that described patient- health education booklet for NCDs (3); studies that described electronic health record implementation (21) and studies that described facility-based records (12).

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Characteristics of included studies

Studies were from Zambia (n=1)[36], Lesotho (n=1)[37], South Africa (n=2)[38 39], China (n=1)[40], and Mongolia (n=1)[41] (Table 3). The designs of included studies were mixed methods (n=1)[41], observational (n=2)[36 37] and qualitative (n=3)[38-40]. Three studies[37 40 41]evaluated PHRs that were implemented and routinely used by the health department in those countries and one study evaluated PHR using diabetes mellitus and hypertension as tracer conditions for NCDs. All the studies described the use of PHRs in outpatient settings, while none measured changes in health outcomes following the use of PHRs.

Quality appraisal

Of the six studies included, five were judged to be of 'medium quality' and one was of 'low quality' (Supplementary file1, S2). For the three qualitative studies[38-40], the methods and data collection were appropriate to answer the research question. However, the studies did not mention the use of a specific method of data analysis or provide a rationale for using an analytic method. Therefore, coherence between data sources, collection, analysis and interpretation could not be fully assessed. The single mixed methods study[41] did not adequately explain the method of qualitative data analysis and was marked down on that criterion. Of the two cross-sectional studies, one[36] provided no information on the sampling strategy and hence it was not possible to make judgements on the representativeness of the sample. Further, no information was reported regarding the non-response rate. The other cross-sectional study provided information on sampling strategy, a reasonable description of the target population and response rate. However, the study was marked down due to insufficient information on measurement variables.[37]

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253 Table 3: Characteristics of included studies

Author, year, LMIC	Study design, sample size	Study objective	Settings	Participants	PHR description	Results
Van der Hoek [36], 1995, Zambia	Cross sectional (n=176,n=184)	To describe the views of patients on the newly introduced health passports by health management in Sesheke district, Zambia	Household surveys (n=2), conducted in the catchment areas of one hospital and two health centres.	Patients	Hardcover booklet with 32 pages known as a health passport, size-14.5*10cm, Demographic information on the cover page. Referral letters and discharge letters are replaced with health passports.	68% of the respondents bought a health passport within 4 months of their introduction and 91% within 16 months. 36% (64/176) and 30% (55/184) of the respondents felt that the passports were expensive.
Henbest[37], 1995, Lesotho	Cross sectional (n=878) Patients(n=691) Nurse clinicians (n=74) and doctors(n=104)	To inform the recommendations for the use of PHR in healthcare services in changing Africa, using the PHR in Lesotho.	Healthcare Centres and household survey	Patients, nurses and doctors	PHR, in the form of a health booklet about the size of a passport (10.5*15cm) with a brightly coloured plasticised cardboard cover. The front cover contains the patient's name, address, date of birth and instruction on the record's use. The back cover has the common symptoms and signs of tuberculosis. The inside printed pages contain forms for recording information such as immunisations, previous significant history drug sensitivity and screening activities. There are 16 blank pages for HCP notes. When	89% of patients preferred to have a PHR than a facility based record. 79% of doctors and 70% of nurses rated the use of PHR as excellent. Doctors and nurses perceived that having a PHR reduces unnecessary repetitions of tests and treatments. They also felt that availability of patient medical information from other HCPs leads to less errors.

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					one record is full, it is stapled to the next record.	
Norden[38],2004 South Africa	Qualitative study Focus group discussions with patients (n=24)	To understand the patients' thoughts on the PHR introduced. It was introduced due to the difficulty in retrieval of medical information at HCP visits from facility based records.	Primary care clinic	Patients	Pocket size booklet with 10 pages (14*8 cm) Demographic information, problem list and allergies recorded in the first page. Blood pressure, blood glucose, body mass index, peak expiratory flow rates are recorded in the subsequent visits. Prescribed treatment and follow up dates are also documented.	Patients felt that the records function as 'medical identification' and are useful in an emergency, provide important background information, and that they enable continuation of care at other facilities. Patients reported that PHRs could motivate them to act on the advice given, and the records also served as a reminder to take their medication and about the dates of follow-up visits.
Kerry[39] , 2005, South Africa	Quality assurance methods with Focus group discussions (n=13) number of participants not stated	To assess, document and improve the PHR System in the Emtshezi Subdistrict, South Africa.	Health care facilities at sub-district level	Patients, nurses, doctors, administrators and pharmacists	Proposed better PHR: A single booklet PHR for every patient in the district. Size – 10 x 21cm named as Health Book; at least 32 pages for clinical notes; a problem list; pages for laboratory and X-ray results and incorporation of the Health Card for Women, TB Card and Antenatal Card. The booklet should have a plastic cover.	Patients and HCPs reported that communication of clinical information between health facilities was poor. HCPs described there were difficulties in retrieval of information and recording at hospitals, the current PHRs were unstructured and multiple PHRs were brought by patients. Patients and HCPs felt that having a single, common PHR to be the definitive ambulatory health record for every patient at district level. Feedback from the FGDs were used to design a better PHR.

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Chen[40] et al, 2014,China	Qualitative study (semi-structured interviews and observations Doctors (n=4), patients(n=13), family members (n=4)	To examine and explore practices surrounding PHR in a setting where medical records have long been managed by patients: the Chinese healthcare system.	Out-patient departments of hospital.	Patients, family members and doctors	Pocket size book which had chief complaint and history handwritten by the clinician. Diagnosis and prescription are documented on the electronic health system and printouts of the same are provided for the patients.	The findings suggest that through engaging in practices of managing and sharing records, patients were able to obtain familiarity with their own records and to provide necessary assistance to locate information for HCPs to use at the point of care. Clinicians felt that a verbal history by the patient was insufficient, as was a history based solely on their own clinic's records.
Ibrahim[41] et al, 2019, Mongolia	Mixed methods, (n=395)	To describe Mongolia's universal patient-held health booklets and their use, explore patients' views on using them and explore training and protocols using tracer conditions diabetes and hypertension.	Out-patient departments of two public hospitals	Patients,	Booklet with 40 pages (14.8*21cm). Demographic information on the cover page, history or handover information in subsequent pages.	94% of patients had PHRs with them at OPD visits. Provision of written information for the three categories of key information items (i.e. diagnosis, prescription/management and follow-up information) for the consultation was low at around 37% in the PHR;- 61% had notes related to medication required and 40% contained information about follow-up. Patients (316, 80.0%) reported consulting their PHRs and associated 'documents' at home. They reported using them for understanding their own condition and communicating it to others

Legends: PHR=patient-held health record, HCP=healthcare HCP, FGD=focus group discussion

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Synthesis of results from included studies

Overall, the included studies were inadequate to demonstrate a clear benefit of using PHRs to improve information availability to HCPs and patients. However, based on the patients' perspective, PHRs were beneficial to the management of their own care. Further, based on the HCPs' perspective PHRs improved the availability of key clinical information for providing care.

Patients carrying their PHRs

Overall, most patients carried their PHRs to healthcare visits. Of the six studies, two reported data on the prevalence of patients bringing their PHR to subsequent visits. In Mongolia, Ibrahim et al[41] reported that 94% (373/395) of chronic NCD patients brought their PHR to their outpatient visits. This was measured by examining the documents brought to the visit by the patients.

In Lesotho, Henbest et al[37] used a survey with patients and HCPs to report on the practice of carrying records to consultations and HCPs' opinions on the availability of PHRs at consultations. Patients' self-reported practice of carrying PHRs to healthcare consultations was 71% (n=672) and HCPs self-reported the availability of the PHR at visits as 62% (doctors, n=81) and 59% (nurse clinicians, n=68) respectively.

In China, Chen et al [40] used interviews with, and observation of, patients to report patients' perspectives on, and patterns of, carrying records to consultations. They found that the patients with chronic NCDs had different PHRs, according to particular conditions/diseases, and clinics or hospitals. Patients often carried only specific PHRs to consultations based on the condition/HCP/clinic they visit.

HCPs' recording information and the quality of information recorded

The HCPs' recording in PHRs was sub-optimal. Of the six studies, only one (from Mongolia[41]) evaluated HCPs' documenting of information in PHRs and the quality of the information recorded for patients with diabetes and hypertension. The overall written information, across three categories

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(diagnosis, prescription and follow-up), within the PHR after the consultation was 37% (n=367). The completeness of PHRs with respect to medication and follow up information were 61% and 40%, respectively. No studies reported the clarity, legibility or accuracy of the recorded information.

Utility of PHR to HCPs

Generally, HCPs perceived PHRs to be useful in recording patients' medical information. In Mongolia, Ibrahim et al[41] did not formally assess HCPs' perspectives on the use of PHR, but they did find that HCPs in the two public outpatient hospitals wrote in the booklets in spite of also having to enter information in other electronic record systems. Further, more than three-quarters (77.4%, n=106) of patients reported that private HCPs they had visited recorded clinic notes in the PHR.

In Lesotho, Henbest et al[37] described HCPs' satisfaction with using PHRs. This was measured in terms of preference for PHRs, practicality (size, durability and confidentiality) and comparison with facility based records (quality of care, access to patient information). Both doctors and nurse clinicians reported a preference for PHRs over facility-based paper records. The HCPs perceived that having a PHR contributed to improving quality of care, by preventing the unnecessary repetition of tests and treatments (86% of nurse clinicians (n=71) and 88% of doctors (n=89)) and that the availability of medical information from other HCP visits helped in the clinical management of patients (78% of nurse clinicians (n=71) and 84% of doctors (n=89)). The qualitative study by Chen et al [40] from China[40], reported that a patient's verbal history of past medical information was perceived as insufficient for providing effective care by doctors. Further, access to previous documented clinical information was perceived by doctors as essential to make appropriate clinical decisions. The other two studies[38 39] did not report the HCPs' perspectives on PHRs.

Protocols and HCP Training

The studies reported very limited protocols and training available for HCPs on the use of PHRs. The study from Mongolia[41] explored the role of protocols and policy directions, with a Government order reinforcing the use of PHRs. However, no written guidelines or training materials regarding the use of

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PHRs or training for clinical handover were reported by HCPs. Another study reported one day training for health workers, with practical instructions on how to use the PHR.[36]

Health outcomes

The included studies do not report measuring changes in health outcomes such as improvement in blood pressure, blood glucose or patients receiving foot examinations or health care advice following the use of PHR.

Utility of PHRs for patients

All included studies reported some use of the information in the PHRs by patients and in general, found that patients regarded PHRs as important documents containing their own medical information. Ibrahim et al[41] described how Mongolian patients (316, 80.0%) consulted their PHRs at home. Patients reported that written information from doctors was important for their own understanding of their condition. Van der Hoek et al[36] reported that some Zambian patients (n=176, 16% (at 6 months) and n=183, 14% (at 18 months)) preferred to keep their records at the clinic itself, due to a fear of damaging the records at home but they did not report whether the patients used the PHRs at home or not. Henbest et al[37] found that most (n=691, 89%) of Lesotho (South African) patients preferred to have a PHR and perceived the record as belonging to them. They reported that having a PHR was important to making information available to the other HCPs they visited. They highlighted that this was very useful in case of travel and emergencies. Norden et al [38] also reported similar findings for South African patients. Patients viewed their PHR as a document containing their own medical information and a tool for continuation of care between HCPs. Additionally, the PHR helped these patients and their families to remember how to manage their own care at home. In their qualitative study, Chen et al [40] found that Chinese patients regarded PHRs as useful. They reported that patients read their documents, organised them chronologically and brought them to each subsequent visit.

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DISCUSSION

Summary of findings

This systematic review identified that there is limited literature from LMICs on the use of PHRs for improving information availability to HCPs and patients for handover communication. This paucity of published research limits our ability to draw conclusions on the benefits or challenges of PHRs in the care of patients with NCDs. Importantly, none of the studies included in the review investigated, reported or evaluated changes in health outcomes after using PHRs in LMICs. Consistent with studies of maternal and child PHRs[42-44], in general, patients brought their PHRs to each clinical consultation. PHRs being in the form of a booklet in Mongolia and China and attaching medical reports (in the form of papers) to the booklet, both reduced the risk of losing them. Although only one study addressed these, the inadequate completeness of the information documented by HCPs, and a lack of protocols and training on the use of PHRs, are both likely to be prominent issues in all the settings as evidenced by the wider literature on training and protocols for medical records documentation.[45-47] The PHRs were perceived as useful by HCPs for documenting the medical history of patients, thus improving the availability of medical information. From the patient perspective, the PHR functioned as a reminder for medication, improved self-care efficacy and enabled continuity of care by providing relevant clinical information to all HCPs involved in patient care. Consistent findings on the use of PHR in improving self-care for the management of chronic conditions were reported and equally reflected in HIC studies.[27 28]

Implications for research and practice

The findings from the review suggest that PHRs may be particularly useful in health systems where little or no medical record keeping occurs at outpatient level, where record retrieval is difficult, or where multiple HCPs may be involved in a person's care for chronic conditions. This is especially important in LMICs as most have pluralistic health systems and no 'gate-keeping', so that patients often change their HCPs, or shop around, and receive care at both public and private facilities.[15 16 45] Patients' provision of a verbal medical history as the only means of transferring previous clinical

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information during HCP visits is, however, insufficient. In LMICs patients' communication of past medical history may be adversely affected by a lack of health literacy, inability to articulate the clinical procedures received, and a lack of confidence against a background of profound power differentials between patients and HCPs.[48-50]

A well-completed PHR may in principle improve medical information availability for HCPs. However, no trials, which assessed the availability of medical information for HCPs, were reported from LMICs. In Germany, Straßner et al[51] conducted a trial investigating the utility of PHRs for asylum seekers in reception centres for improving medical information availability for HCPs. The trial found that a PHR for asylum seekers increased the availability of health information for HCPs (aOR 4.22, 95% CI: 2.64 to 6.73), reduced missing information (aOR 0.89, 95% CI: 0.42 to 1.88), and reduced HCP dissatisfaction (aOR 0.43, 95% CI: 0.16 to 1.14) .[51] Previous HIC studies have shown that the use of PHRs for patients with diabetes and hypertension have led to decreases in HbA1c ($p<0.001$)[52-54], decrease in diastolic blood pressure ($p<0.05$) [52], and improvement in receiving foot examinations (OR: 1.68; 95% CI 1.12 to 2.50)[54], having physical activity or exercise advised (OR: 1.84; 95% CI 1.16 to 2.92) [54], and monitoring of weight, blood glucose and cholesterol[52]. Similar results may be achieved in LMICs for monitoring blood pressure/glucose/cholesterol, improving foot/eye examination if PHRs are well documented.

Patients, especially those with chronic diseases, usually carry the records they have to HCP visits, especially if they know HCPs will use them. [15 16 45 55] However, in the absence of an organised PHR, patients either bring very little information, such as a prescription alone (common in LMICs), or they bring all the previous pieces of paper that form their home held medical records, making it difficult for HCPs to find the relevant information.[16] Many chronic disease patients have multi-morbidities and often have numerous papers or clinic specific PHRs from the different clinics and hospitals they visit, but they often only carry notes from previous visits to the clinic they are going to visit.[28] Thus, the HCPs do not receive the information about medication and management plans produced by

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other doctors in other clinics or hospitals. Therefore, it is necessary to develop generic/universal and pragmatic PHRs and create awareness of the importance of using a single generic PHR for all healthcare visits, irrespective of the clinic/hospital, so that all HCPs have the necessary medical information available to make the best clinical decisions.

The WHO recommends that ideally a PHR should be used as complementary to a facility based medical record.[56] Prior reviews from HICs showed that, when HCPs had to enter information in facility based records (electronic or paper based) and PHRs, this multiple recording was an unwelcome burden.[57 58] However, documented discharge information for patients /families (e.g., summaries/letters) from HCPs has been found to reduce post-discharge complications[59] and re-admissions,[60] and to improve patient satisfaction[61] and health outcomes.[62] Arguably, it would more efficient and accurate if there were one record system accessible and useful to both patients and HCPs.

Many LMICs are progressing with the implementation of electronic health records, which may enable health systems to overcome difficulties in retrieving information from paper-based facility-based medical records.[63 64] Despite such advancements in technology and information technology capability in some middle-income countries, paper records will remain the prominent form of medical records and are unlikely to be completely replaced by electronic records at least for the near future. Furthermore, the issues of integrating electronic health records across different levels of care in health systems and public-private information exchange are often not addressed in existing health information management systems.[65] Therefore, PHRs having a minimum of data such as diagnosis, medication, lab results and follow-up will enable better information transfer in the health systems transitioning to electronic health records.

Strengths and limitations

Our systematic review has several strengths and limitations. The review protocol was registered at PROSPERO and we followed PRISMA guidelines in our systematic review. A comprehensive search

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strategy was employed to locate studies from LMICs. Furthermore, our search from the grey literature and discussion with experts in the field helped us to identify generic PHRs from Malawi,[66] Namibia[67] and South Africa.[68]However, these PHRs have not been evaluated systematically, and we have limited information on the usefulness of these records. However, our study also has some limitations. Given the small number of relevant articles, we found very few RCTs on which to base conclusions of robust experimental findings. Scarce data on generic PHRs from LMICs limit our ability to interpret the usefulness of these records in improving health outcomes of patients with NCDs.

CONCLUSION

Available data on the use of non-maternity PHR for handover communication between HCPs and HCPs with patients in LMICs are sparse. Existing studies on PHR in LMICs are of medium to low quality, and heterogeneous in terms of study design, population and context. Further, health outcome assessments after the introduction of PHRs are not available. The available studies show that patients value PHRs for their own use and are likely to carry them to HCPs. HCPs, in turn, consider that PHRs enhance documented information transfer from previous visits specifically when there is difficulty in retrieving past medical information. With LMICs progressing with electronic health record implementation, the role of paper based PHRs in overcoming issues of integration and for improving adherence to effective self-care needs re-evaluating. Future research should explore minimum information requirements for major conditions, the format and types of PHRs of most utility to patients and HCPs (electronic or paper-based), and barriers and facilitators to effective use of PHRs across all HCPs for any one patient. Further, the content and HCP training and protocols need to be developed and evaluated for the effective scale-up of interventions to promote the effective use of PHRs for continuity of care and patient self-care.

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Declaration of interest

We declare no competing interest.

Author contributions

JLM, PJ, SG and SMH conceived the article. JLM performed the literature search. JLM, BD, JP collected data from individual studies and interpreted the data. JLM wrote the first draft of the paper. SMH, SG, PJ, AL and CH revised the first and subsequent drafts. All authors contributed to interpretation of the findings and revised the manuscript for important intellectual content.

Data sharing

No additional data available

Fig 1. Legends

LMIC=Low and middle income countries, CVD=Cardiovascular disease, PHR=Patient-held health record, NCD=Non-communicable disease

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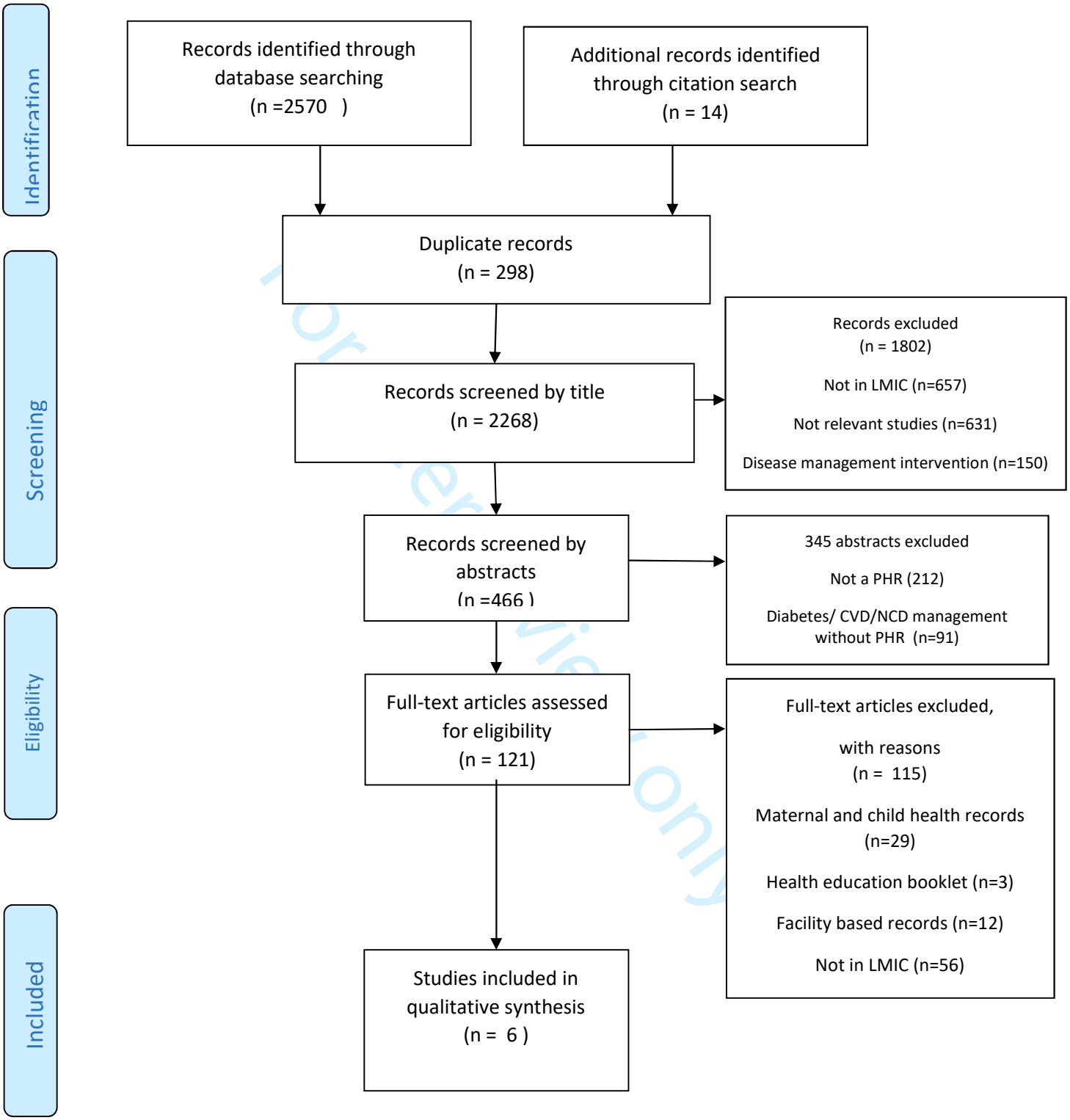


Fig 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram

SUPPLEMENTAL MATERIAL

S1.Search strategy in EMBASE

Patient-held records

1. (exp medical records/or exp medical records, problem-oriented/or exp medical records systems, computerized/or exp nursing records/or (((medical or health) and record*).mp. or (case note* or case record*).mp.) or (record* or book* or handbook* or card*).mp AND ((held adj3 (patient or parent or person or woman or man or family or consumer)).mp. or (carried adj3 (patient or parent or person or woman or man or family or client or consumer)).mp.)) OR ((log-book* or logbook*).mp. or home based record*.mp. or passport*.mp. or personal health record*.mp.)
2. ((patient-held or patient-held or home based or home-based or personal child or "mother and child" or "maternal and child" or mother-child) adj2 (record* or book* or handbook* or card*)).mp
3. ((patient access or personal or patient-held) adj2(record*or portal * or health record* or information* or prescription* or electronic record*)).mp
4. 1 or 2 or 3

Low-and Middle-income countries

5. exp developing country/ or exp medically underserved/ or developing countr$.mp. or medically underserved area$1.mp. or low income countr$.mp. or middle income country.mp. or low resource.mp. or resource poor.mp. or global.mp. or exp Africa/ or exp "South and Central America"/or exp asia/ or exp Caribbean islands/ or exp pacific islands/ or exp eastern Europe/ or exp Indian Ocean/or south america$1.mp. or Africa$1.mp. or Caribbean.mp. or central America$1.mp. or south America$1.mp. or eastern Europe$1.mp. or pacific island$.mp. or Indian ocean island$.mp. or asia.mp. or Afghan$.mp. or Bangladesh$1.mp. or Benin$.mp. or Burkina Faso.mp. or Burkinabe.mp. or Burundi$.mp. or Cambodia$1.mp. or Central African.mp. or Chad$.mp. or Comor$.mp. or Congo$.mp. or Eritrea$1.mp. or Ethiopia$1.mp. or Gambia$1.mp. or Guinea$1.mp. or Haiti$.mp. or Kenya$1.mp. or Korea$1.mp. or exp North Korea/ or Kyrgyz$.mp. or Liberia$1.mp. or Madagascar.mp. or Malagasy.mp. or Malawi$.mp. or mali$.mp. or mozambi$.mp. or Myanmar$.mp. or Nepal$.mp. or Niger$.mp. or Rwanda$1.mp. or Sierra Leone$.mp. or Somalia$1.mp. or Tajik$.mp. or Tanzania$1.mp. or Togo$.mp. or Uganda$1.mp. or Zimbabwe$.mp. or Angola$1.mp. or Armenia$1.mp. or Beliz$.mp. or Bhutan$.mp. or Bolivia$1.mp. or Cameroon$.mp. or Cape Verde$.mp. or Congo$.mp. or "Côte d'Ivoire".mp. or Ivory Coast.mp. or Ivorian.mp. or Djibouti.mp. or Egypt$.mp. or El Salvador.mp. or Salvadoran.mp. or Fiji$.mp. or Georgia$1.mp. or Ghana$.mp. or Guatemala$1.mp. or Guyan$.mp. or Hondura$.mp. or Indonesia$1.mp. or India$1.mp. or Iraq$1.mp. or Kiribati.mp. or Kosov$.mp. or Lao$.mp. or Lesotho.mp. or Marshall Islands.mp. or Marshallese.mp. or Mauritania$1.mp. or Micronesia$1.mp. or Moldov$.mp. or Mongolia$1.mp. or Morocc$.mp. or Nicaragua$1.mp. or Nigeria$1.mp. or Pakistan$1.mp. or Papua New Guinea$1.mp. or Paraguay$.mp. or Philippines.mp. or Filipino.mp. or Samoa$1.mp. or sao tome$.mp. or Senegal$.mp. or Solomon Island$.mp. or sri lanka$1.mp. or Sudan$.mp. or Swazi$.mp. or Syria$1.mp. or Timor$.mp. or

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Tonga$1.mp. or Turkmen$.mp. or Tuvalu$.mp. or Ukrain$.mp. or Uzbek$.mp. or Vanuat$1.mp. or Vietnam$.mp. or West Bank.mp. or Gaza.mp. or Yemen$.mp. or Zambia$1.mp. or Albania$1.mp. or Algeria$1.mp. or "Antigua and Barbuda".mp. or antiguan.mp. or barbudan.mp. or Azerbaijan$1.mp. or Belarus$.mp. or Bosnia$1.mp. or Botswana.mp. or Brazil$.mp. or Bulgaria$1.mp. or Chile$.mp. or China.mp. or Chinese.mp. or Colombia$1.mp. or Costa Rica$1.mp. or Cuba$1.mp. or Dominica$1.mp. or Ecuador$.mp. or Gabon$.mp. or Grenad$.mp. or Iran$.mp. or Jamaica$1.mp. or Jordan$.mp. or Kazakhstan$1.mp. or Latvia$1.mp. or Leban$.mp. or Libya$1.mp. or Lithuania$1.mp. or Macedonia$1.mp. or Malaysia$1.mp. or Maldiv$.mp. or mauriti$.mp. or Mexic$.mp. or Montenegr$.mp. or Namibia$1.mp. or Palau$.mp. or Panama$.mp. or Peru$.mp. or Romania$1.mp. or Russia$1.mp. or Serbia$1.mp. or Seychell$.mp. or South Africa$1.mp. or Saint Kitts.mp. or Saint Lucia.mp. or Saint Vincent.mp. or Suriname$1.mp. or Thai$.mp. or Tunisia$1.mp. or Turk$.mp. or Uruguay$.mp. or Venezuala$1.mp

6. 4 and 5

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S2. Table summaries of MMAT quality appraisal assessments for included studies

Author, year, LMIC	Type of study	4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)
Van der Hoek [23], 1995, Zambia	Quantitative descriptive	Can't tell	Can't tell	Yes	No	Yes	2
Henbest[24], 1995, Lesotho	Quantitative descriptive	Yes	Yes	Can't tell	Yes	Yes	4
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
Norden[25],2004 South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Kerry[26] , 2005, South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Chen[27] et al, 2014,China	Qualitative	Yes	Yes	Yes	Yes	Can't tell	4
		4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)

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Ibrahim[28] et al, 2019, Mongolia	Quantitative descriptive	Yes	Yes	Yes	Yes	Yes	5 (Has description of non-responders and confounders)
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
		5.1. Is there an adequate rationale for using a mixed methods design to address the research question?	5.2. Are the different components of the study effectively integrated to answer the research question?	5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?	
	Mixed methods	Yes	Can't tell	Yes	Can't tell	Yes	3(Integration may have occurred in the interpretation of results-It is unclear, cannot tell if there is divergent findings)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1, Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5,7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7,8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file, S1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10,11

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12,13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13 and Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 and Supplementary file, S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	18-20
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Na
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20

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Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	23
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5,7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7,8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file, S1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10,11

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Section/topic	#	Checklist item	Reported on page #
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Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12,13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13 and Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 and Supplementary file, S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	18-20
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Na
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	23
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A systematic review on the use of patient-held health records in low and middle-income countries.

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ABSTRACT

Objective

To review the available evidence on the benefit of patient-held records(PHRs), other than maternal and child health records, for improving the availability of medical information for handover communication between healthcare providers (HCPs) and/or between HCPs and patients in low and middle income countries (LMIC).

Methods

The literature searches were conducted in PubMed, EMBASE, CINAHL databases for manuscripts without any restrictions on dates/language. Additionally, articles were located through citation checking using previous systematic reviews and a grey literature search by contacting experts, searching of the World Health Organisation (WHO) website and Google Scholar.

Results

Six observational studies in four LMICs met the inclusion criteria. However, no studies reported on health outcomes after using PHRs. Studies in the review reported patients' experience of carrying the records to HCPs (n=3), quality of information available to HCPs (n=1), and the utility of these records to patients (n=6) and HCPs (n=4). Most patients carry PHRs to healthcare visits. One study assessed the completeness of clinical handover information and found that only 41% (161/395) of PHRs were complete with respect to key information on diagnosis, treatment and follow-up. No protocols or guidelines for HCPs were reported for use of PHRs. The HCPs perceived the use of PHRs improved medical information availability from other HCPs. From the patient perspective, PHRs functioned as documented source of information about their own condition.

Conclusion: Limited data on existing PHRs make their benefits for improving health outcomes in LMICs uncertain. This knowledge gap calls for research on understanding the dynamics and outcomes of PHR use by patients and HCPs and in health systems interventions.

Keywords -**International health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH**

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Study design that enabled the accumulation and presentation of current evidence on the use of patient-held health records other than maternal and child health records in low and middle-income countries.
- Thorough literature search of three major electronic databases and reporting as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.
- Inconclusive results due to no evidence from trials.
- Small number of studies and no reporting on health outcomes, making it difficult to determine usefulness for patients with non-communicable diseases.

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INTRODUCTION

To meet sustainable development goals[1] and deliver quality care[2], health systems in low and middle income countries (LMICs) need to be redesigned and strengthened.[3] In line with this agenda, the World Health Organization (WHO) has formulated a health systems framework that describes six building blocks for health systems; of these, good service delivery comprises the delivery of effective, safe, quality health care and continuity of care is an important characteristic for delivering quality care.[4]

In many LMICs, healthcare delivery is usually organised around acute and single episode care, from different facilities and health care providers (HCP).[5] Such a model of healthcare delivery is inadequate due to increasing non-communicable disease (NCD) burden in LMICs.[6] Continuity of care needed for efficient and effective care, is simply defined as "the seamless provision of healthcare between settings and over time".[7] Informational continuity represents the cornerstone of continuity of care and is defined as the use of information on past events and personal circumstances to make current care appropriate for each individual.[8] Handover communication not only relates to the process of passing patient-specific health information between HCPs, from one visit to another, but also from HCPs to the patient and family for ensuring patient care-continuity and self-management.[9]

Complete and accurate clinical information about previous management and treatment is necessary for managing all patients but especially patients with chronic conditions due to a need for on-going care from a range of HCPs.[10] Inadequate documented handover for HCPs often results in adverse events[11 12] or increased in-patient visits[13], repeated tests or examinations leading to costly healthcare, burdensome to patients and a waste of resources.[13] The ability to view patients' medical records, across healthcare visits and facilities helps to facilitate health information exchange.[14] The absence of out-patient medical records,[15 16]difficulties in retrieval of facility based records for out-

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patient visits,[17] and lack of integrated patient electronic medical records between departments and between health facilities in LMICs contribute to poor informational continuity for HCPs.[18]

A patient-held health record (PHR) can be a viable solution for improving sharing of documented medical information across HCPs.[19] For the purpose of this review, PHRs are formal records given to patients in the form of booklets, which contain patients' medical information. Our hypothesis is that such PHRs with documented medical notes from HCPs can be used as a tool for improving medical information availability for HCPs.[15 20]

Paper-based PHRs are used extensively in maternity care, as home-based records or maternal and child health handbooks for sharing information across antenatal, labour and post-natal care by all HCPs even in LMIC settings.[21] Three systematic reviews on maternity and child records indicate that PHRs improve HCP-patient communication and facilitate referrals, and both HCPs and women give positive feedback on their use.[22-25]

There are several systematic reviews[19 26-29] of PHRs for chronic conditions from high income countries (HIC), which evaluated their usefulness in cancer care,[28] and chronic conditions,[26 27] and for people with severe mental illness[29] or undocumented immigrants.[19] These condition-specific PHRs were designed to mainly improve communication between HCP and patients.[30] The reviews found that PHRs may improve a patient's sense of control and empowerment, leading to better involvement in their own care.[28 29]

There is no systematic review of PHRs for conditions other than maternal and child health from LMICs, while there is evidence that generic PHRs do exist in some LMICs.[31] A study in India found that patients were carrying unstructured sheets of paper to out-patient visits and patient-held health booklets were an acceptable and pragmatic intervention to improve information exchange for chronic NCD patients by patients and HCPs.[15]

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The current review aimed to investigate the benefits of generic PHRs used in LMICs to ensure the availability of medical information for clinical handover communication and continuity of care between HCPs or outpatient health facility visits, with or without their use for communication between HCPs and patients. This review investigated non-maternal-child health PHRs, designed to record patient histories and healthcare information across a range of conditions and HCPs, including primary and secondary care and not specific to one facility or one speciality.

METHODS

The protocol for this review is registered with the PROSPERO International Prospective Register of Systematic Reviews (CRD42019139365)[32].

Selection criteria

The published literature was searched for quantitative randomised controlled trials (RCTs), cluster RCTs, quasi-experimental studies, cross sectional and pre-post study designs, qualitative and mixed-method studies. Case reports, commentaries and review articles were excluded studies. The eligibility criteria are provided in Table 1.

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160

161 Table 1 Inclusion and exclusion criteria

Title	Inclusion criteria	Exclusion criteria
Population	• Participants (patients, carers and/or health care HCPs) from any health background, of 18 years or more, using a PHR were considered. • All healthcare settings within LMICs (as defined by the World Bank 2018)[33] were considered.	• Children and pregnant women • Patients/ carers/HCPs from HIC
Intervention	• Studies which include design, implementation or evaluation of formalised PHRs , with the purpose of improving information exchange and communication between visits to the same facility for HCPs, across different health care facilities (primary to secondary/vice versa), and documented information from HCPs to patients (for patients' own care). • Studies which describe PHRs which are clinically focused and person specific or generic PHRs. Studies which describe condition specific PHRs, they should include NCDs such as Diabetes Mellitus, hypertension, cardiovascular diseases, chronic respiratory diseases or cancer. • PHRs are defined as any formal medical document in the form of a booklet held by patients, which can be used across health care settings or visits, contain patient histories and healthcare information to guide healthcare workers providing care.	• Facility based medical records, which are for HCP use. • Home based records used for maternal and child health care. • Patient-held diaries used for monitoring values such as home based blood pressure monitoring booklets or patient instructional booklets for example for diabetic foot care, or patient-held records for single communicable disease such as tuberculosis. • Studies that focus on record use for specific and isolated transitions of care, such as hospital shift-change or discharge. Interventions such as a discharge summary or referral letter alone. • Studies that focus on a single function of patient-held medical records such as medication prescriptions, lab results, blood pressure or blood sugar monitoring or out-patient registration papers.
Outcomes	• Outcome relevant to the quality of verbal and/or documented patient-specific information and	

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	communication. Standards of information and recording (completeness, accuracy and clarity) of the documented handover information was included. • Patients' and HCPs' views on how PHRs enabled/did not enable communication and documented information exchange was also included. Patients carrying records to consultation, availability of documented medical information for HCPs during consultation was also included. • Patient and HCP satisfaction with PHR, patient centred communication outcomes such as patient satisfaction, recall, understanding and adherence have been included. • Clinical outcomes and adverse events associated with quality of handover communication were also of interest, including (but not limited to) readmissions, diagnostic delays, healthcare utilisation or improved appointment rates, and death. Intermediate outcomes such as blood pressure/blood glucose monitoring, medication management/reconciliation and/monitoring of lab values were included.	• Descriptive studies which describe the distribution of PHRs without information on outcomes of PHRs.

Legend: PHR=patient-held health record, LMIC=low and middle income countries, HIC=high income countries, HCP= health care HCP NCD=non-communicable diseases

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Search strategy

The search strategy (Supplementary file 1, S1) was developed for the electronic databases MEDLINE (Ovid) to identify journal articles. The search strategy was made following previous reviews regarding PHRs[26 28] EMBASE (Ovid EMBASE Classic + EMBASE (1947 to present)) and CINAHL (via EBSCO (1980 to present) were searched for studies reporting PHR in LMICs without a librarian. Additionally, citations were hand searched using relevant systematic reviews[19 28] to locate further relevant studies. Key search terms related to PHRs and LMICs were used to build the search strategy. The search terms used for PHRs were "patient-held book* or handbook* or card", "patient passport", "log book" and "home based records". Searches in the electronic databases were conducted from database inception until September 2020. No limits on language or publication year were applied during the search. Grey literature search was done by contacting authors of papers, which reported the use of patient-held records, discussion with experts and using the website of World Health Organisation (WHO) and google scholar.

Study selection

The search results were uploaded to the reference management software Zotero to remove duplicates. Two reviewers (LJ and DB) independently screened the remaining studies' titles. The initial title screening was broad and retained all studies that referred to PHRs. Two reviewers (LJ and DB) independently screened the abstracts and full texts against the inclusion criteria. Any uncertainty surrounding the inclusion of a study or disagreement following discussions between reviewers were resolved through the assessment of an additional reviewer (SMH).

Data extraction

Two reviewers (DB and PJ) independently extracted data from the included studies. LJ reviewed all the data extraction tables. If there were any discrepancies during this sampling check, discussion took place and for final clarification a third, impartial reviewer (JP or SMH) was consulted.

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Data were extracted from published studies using a data extraction form in Microsoft Excel and included details of authors, study settings and country, the objectives and features of the PHR, use of PHRs, and outcomes.

Quality assessment

Quality assessment was done by two reviewers (LJ and DB) using the Mixed Methods Appraisal Tool (MMAT).[34] This provided an overall description of the quality of studies and used descriptors 'low', 'medium' and 'high' to provide an indication of the quality of the included studies. For quantitative, qualitative and cross-sectional studies, all five criteria needed to be met to be classified as 'high quality'. Studies that met three to four criteria were classified as 'medium quality' and studies that met one or two criteria were regarded as 'low quality'. For mixed-methods studies, the overall score was dependent on the lowest score of each of the study components (qualitative and quantitative); therefore, scores were determined by the quality of the weakest component. In case of disagreements on quality assessment, the final decision was taken after discussion with an independent third reviewer (JP or SMH).

Data analysis

Given the diversity of populations, settings, the non-uniformity of interventions, and the variations in outcome measures, a statistical meta-analysis was not appropriate. Hence, the review used a narrative synthesis. Papers in the review have been summarised descriptively as follows (Table 2).

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206 Table 2: Description of results

Result	Description of the result
Patients carrying the PHRs to HCP visits	Presented as frequencies or number of patients carrying the records to visits or as prevalence of written clinical information availability for HCPs at visits.
HCPs' recording of information and Quality of information recorded	HCPs' recording of information in the PHR is presented as information availability for patients on leaving the facility or patient/HCP self-reported availability of documented information. The quality of information recorded for this review is defined as the completeness of the information on key elements such as diagnosis, medication/treatment details (including lab values) and follow-up information, clarity or legibility of the information recorded and accuracy of the information being recorded based on treatment guidelines (For example; mismatch of diagnosis and treatment prescribed)/comparison of data with facility based records. Completeness data are reported as frequencies and stand-alone or comparative data based on key missing components.
Utility of PHR to HCPs and patients	For the review, utility as perceptions of patients and HCPs about using PHRs, satisfaction with use, usability in terms of ease of reading the records, and the functions they serve (for information exchange, clinical information recording at each HCP visit, patient education or as an aide memoire for patients).

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207

208 Legend: PHR=patient-held health record, HCP=health care HCP

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Patient and public involvement

This research was done without patient or public involvement.

Ethics Approval

No ethical approval was needed because data from previous published studies is being used.

RESULTS

Study selection

The findings of the search strategy are summarized in Fig 1 as a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart.[35] 2286 titles were screened and 466 abstracts were included. 121 full papers were reviewed and six studies were included in the final review. The references of included studies were reviewed for potential studies and no new studies were included. We excluded 115 articles because: studies that described PHRs but were not conducted in LMICs (56); studies that described maternal or child health records in LMICs (29); papers that described patient- health education booklet for NCDs (3); studies that described electronic health record implementation (21) and studies that described facility-based records (12).

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Characteristics of included studies

Studies were from Zambia (n=1)[36], Lesotho (n=1)[37], South Africa (n=2)[38 39], China (n=1)[40], and Mongolia (n=1)[41] (Table 3). The designs of included studies were mixed methods (n=1)[41], observational (n=2)[36 37] and qualitative (n=3)[38-40]. Three studies[37 40 41]evaluated PHRs that were implemented and routinely used by the health department in those countries and one study evaluated PHR using diabetes mellitus and hypertension as tracer conditions for NCDs. All the studies described the use of PHRs in outpatient settings, while none measured changes in health outcomes following the use of PHRs.

Quality appraisal

Of the six studies included, five were judged to be of 'medium quality' and one was of 'low quality' (Supplementary file1, S2). For the three qualitative studies[38-40], the methods and data collection were appropriate to answer the research question. However, the studies did not mention the use of a specific method of data analysis or provide a rationale for using an analytic method. Therefore, coherence between data sources, collection, analysis and interpretation could not be fully assessed. The single mixed methods study[41] did not adequately explain the method of qualitative data analysis and was marked down on that criterion. Of the two cross-sectional studies, one[36] provided no information on the sampling strategy and hence it was not possible to make judgements on the representativeness of the sample. Further, no information was reported regarding the non-response rate. The other cross-sectional study provided information on sampling strategy, a reasonable description of the target population and response rate. However, the study was marked down due to insufficient information on measurement variables.[37]

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254 Table 3: Characteristics of included studies

Author, year, LMIC	Study design, sample size	Study objective	Settings	Participants	PHR description	Results
Van der Hoek [36], 1995, Zambia	Cross sectional (n=176,n=184)	To describe the views of patients on the newly introduced health passports by health management in Sesheke district, Zambia	Household surveys (n=2), conducted in the catchment areas of one hospital and two health centres.	Patients	Hardcover booklet with 32 pages known as a health passport, size-14.5*10cm, Demographic information on the cover page. Referral letters and discharge letters are replaced with health passports.	68% of the respondents bought a health passport within 4 months of their introduction and 91% within 16 months. 36% (64/176) and 30% (55/184) of the respondents felt that the passports were expensive.
Henbest[37], 1995, Lesotho	Cross sectional (n=878) Patients(n=691) Nurse clinicians (n=74) and doctors(n=104)	To inform the recommendations for the use of PHR in healthcare services in changing Africa, using the PHR in Lesotho.	Healthcare Centres and household survey	Patients, nurses and doctors	PHR, in the form of a health booklet about the size of a passport (10.5*15cm) with a brightly coloured plasticised cardboard cover. The front cover contains the patient's name, address, date of birth and instruction on the record's use. The back cover has the common symptoms and signs of tuberculosis. The inside printed pages contain forms for recording information such as immunisations, previous significant history drug sensitivity and screening activities. There are 16 blank pages for HCP notes. When	89% of patients preferred to have a PHR than a facility based record. 79% of doctors and 70% of nurses rated the use of PHR as excellent. Doctors and nurses perceived that having a PHR reduces unnecessary repetitions of tests and treatments. They also felt that availability of patient medical information from other HCPs leads to less errors.

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					one record is full, it is stapled to the next record.	
Norden[38],2004 South Africa	Qualitative study Focus group discussions with patients (n=24)	To understand the patients' thoughts on the PHR introduced. It was introduced due to the difficulty in retrieval of medical information at HCP visits from facility based records.	Primary care clinic	Patients	Pocket size booklet with 10 pages (14*8 cm) Demographic information, problem list and allergies recorded in the first page. Blood pressure, blood glucose, body mass index, peak expiratory flow rates are recorded in the subsequent visits. Prescribed treatment and follow up dates are also documented.	Patients felt that the records function as 'medical identification' and are useful in an emergency, provide important background information, and that they enable continuation of care at other facilities. Patients reported that PHRs could motivate them to act on the advice given, and the records also served as a reminder to take their medication and about the dates of follow-up visits.
Kerry[39] , 2005, South Africa	Quality assurance methods with Focus group discussions (n=13) number of participants not stated	To assess, document and improve the PHR System in the Emtshezi Subdistrict, South Africa.	Health care facilities at sub-district level	Patients, nurses, doctors, administrators and pharmacists	Proposed better PHR: A single booklet PHR for every patient in the district. Size – 10 x 21cm named as Health Book; at least 32 pages for clinical notes; a problem list; pages for laboratory and X-ray results and incorporation of the Health Card for Women, TB Card and Antenatal Card. The booklet should have a plastic cover.	Patients and HCPs reported that communication of clinical information between health facilities was poor. HCPs described there were difficulties in retrieval of information and recording at hospitals, the current PHRs were unstructured and multiple PHRs were brought by patients. Patients and HCPs felt that having a single, common PHR to be the definitive ambulatory health record for every patient at district level. Feedback from the FGDs were used to design a better PHR.

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Chen[40] et al, 2014,China	Qualitative study (semi-structured interviews and observations Doctors (n=4), patients(n=13), family members (n=4)	To examine and explore practices surrounding PHR in a setting where medical records have long been managed by patients: the Chinese healthcare system.	Out-patient departments of hospital.	Patients, family members and doctors	Pocket size book which had chief complaint and history handwritten by the clinician. Diagnosis and prescription are documented on the electronic health system and printouts of the same are provided for the patients.	The findings suggest that through engaging in practices of managing and sharing records, patients were able to obtain familiarity with their own records and to provide necessary assistance to locate information for HCPs to use at the point of care. Clinicians felt that a verbal history by the patient was insufficient, as was a history based solely on their own clinic's records.
Ibrahim[41] et al, 2019, Mongolia	Mixed methods, (n=395)	To describe Mongolia's universal patient-held health booklets and their use, explore patients' views on using them and explore training and protocols using tracer conditions diabetes and hypertension.	Out-patient departments of two public hospitals	Patients,	Booklet with 40 pages (14.8*21cm). Demographic information on the cover page, history or handover information in subsequent pages.	94% of patients had PHRs with them at OPD visits. Provision of written information for the three categories of key information items (i.e. diagnosis, prescription/management and follow-up information) for the consultation was low at around 37% in the PHR;- 61% had notes related to medication required and 40% contained information about follow-up. Patients (316, 80.0%) reported consulting their PHRs and associated 'documents' at home. They reported using them for understanding their own condition and communicating it to others

Legends: PHR=patient-held health record, HCP=healthcare HCP, FGD=focus group discussion

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Synthesis of results from included studies

Overall, the included studies were inadequate to demonstrate a clear benefit of using PHRs to improve information availability to HCPs and patients. However, based on the patients' perspective, PHRs were beneficial to the management of their own care. Further, based on the HCPs' perspective PHRs improved the availability of key clinical information for providing care.

Patients carrying their PHRs

Overall, most patients carried their PHRs to healthcare visits. Of the six studies, two reported data on the prevalence of patients bringing their PHR to subsequent visits. In Mongolia, Ibrahim et al[41] reported that 94% (373/395) of chronic NCD patients brought their PHR to their outpatient visits. This was measured by examining the documents brought to the visit by the patients.

In Lesotho, Henbest et al[37] used a survey with patients and HCPs to report on the practice of carrying records to consultations and HCPs' opinions on the availability of PHRs at consultations. Patients' self-reported practice of carrying PHRs to healthcare consultations was 71% (n=672) and HCPs self-reported the availability of the PHR at visits as 62% (doctors, n=81) and 59% (nurse clinicians, n=68) respectively.

In China, Chen et al [40] used interviews with, and observation of, patients to report patients' perspectives on, and patterns of, carrying records to consultations. They found that the patients with chronic NCDs had different PHRs, according to particular conditions/diseases, and clinics or hospitals. Patients often carried only specific PHRs to consultations based on the condition/HCP/clinic they visit.

HCPs' recording information and the quality of information recorded

The HCPs' recording in PHRs was sub-optimal. Of the six studies, only one (from Mongolia[41]) evaluated HCPs' documenting of information in PHRs and the quality of the information recorded for patients with diabetes and hypertension. The overall written information, across three categories

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(diagnosis, prescription and follow-up), within the PHR after the consultation was 37% (n=367). The completeness of PHRs with respect to medication and follow up information were 61% and 40%, respectively. No studies reported the clarity, legibility or accuracy of the recorded information.

Utility of PHR to HCPs

Generally, HCPs perceived PHRs to be useful in recording patients' medical information. In Mongolia, Ibrahim et al[41] did not formally assess HCPs' perspectives on the use of PHR, but they did find that HCPs in the two public outpatient hospitals wrote in the booklets in spite of also having to enter information in other electronic record systems. Further, more than three-quarters (77.4%, n=106) of patients reported that private HCPs they had visited recorded clinic notes in the PHR.

In Lesotho, Henbest et al[37] described HCPs' satisfaction with using PHRs. This was measured in terms of preference for PHRs, practicality (size, durability and confidentiality) and comparison with facility based records (quality of care, access to patient information). Both doctors and nurse clinicians reported a preference for PHRs over facility-based paper records. The HCPs perceived that having a PHR contributed to improving quality of care, by preventing the unnecessary repetition of tests and treatments (86% of nurse clinicians (n=71) and 88% of doctors (n=89)) and that the availability of medical information from other HCP visits helped in the clinical management of patients (78% of nurse clinicians (n=71) and 84% of doctors (n=89)). The qualitative study by Chen et al [40] from China[40], reported that a patient's verbal history of past medical information was perceived as insufficient for providing effective care by doctors. Further, access to previous documented clinical information was perceived by doctors as essential to make appropriate clinical decisions. The other two studies[38 39] did not report the HCPs' perspectives on PHRs.

Protocols and HCP Training

The studies reported very limited protocols and training available for HCPs on the use of PHRs. The study from Mongolia[41] explored the role of protocols and policy directions, with a Government order reinforcing the use of PHRs. However, no written guidelines or training materials regarding the use of

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PHRs or training for clinical handover were reported by HCPs. Another study reported one day training for health workers, with practical instructions on how to use the PHR.[36]

Health outcomes

The included studies do not report measuring changes in health outcomes such as improvement in blood pressure, blood glucose or patients receiving foot examinations or health care advice following the use of PHR.

Utility of PHRs for patients

All included studies reported some use of the information in the PHRs by patients and in general, found that patients regarded PHRs as important documents containing their own medical information. Ibrahim et al[41] described how Mongolian patients (316, 80.0%) consulted their PHRs at home. Patients reported that written information from doctors was important for their own understanding of their condition. Van der Hoek et al[36] reported that some Zambian patients (n=176, 16% (at 6 months) and n=183, 14% (at 18 months)) preferred to keep their records at the clinic itself, due to a fear of damaging the records at home but they did not report whether the patients used the PHRs at home or not. Henbest et al[37] found that most (n=691, 89%) of Lesotho (South African) patients preferred to have a PHR and perceived the record as belonging to them. They reported that having a PHR was important to making information available to the other HCPs they visited. They highlighted that this was very useful in case of travel and emergencies. Norden et al [38] also reported similar findings for South African patients. Patients viewed their PHR as a document containing their own medical information and a tool for continuation of care between HCPs. Additionally, the PHR helped these patients and their families to remember how to manage their own care at home. In their qualitative study, Chen et al [40] found that Chinese patients regarded PHRs as useful. They reported that patients read their documents, organised them chronologically and brought them to each subsequent visit.

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DISCUSSION

Summary of findings

This systematic review identified that there is limited literature from LMICs on the use of PHRs for improving information availability to HCPs and patients for handover communication. This paucity of published research limits our ability to draw conclusions on the benefits or challenges of PHRs in the care of patients with NCDs. Importantly, none of the studies included in the review investigated, reported or evaluated changes in health outcomes after using PHRs in LMICs. Consistent with studies of maternal and child PHRs[42-44], in general, patients brought their PHRs to each clinical consultation. PHRs being in the form of a booklet in Mongolia and China and attaching medical reports (in the form of papers) to the booklet, both reduced the risk of losing them. Although only one study addressed these, the inadequate completeness of the information documented by HCPs, and a lack of protocols and training on the use of PHRs, are both likely to be prominent issues in all the settings as evidenced by the wider literature on training and protocols for medical records documentation.[45-47] The PHRs were perceived as useful by HCPs for documenting the medical history of patients, thus improving the availability of medical information. From the patient perspective, the PHR functioned as a reminder for medication, improved self-care efficacy and enabled continuity of care by providing relevant clinical information to all HCPs involved in patient care. Consistent findings on the use of PHR in improving self-care for the management of chronic conditions were reported and equally reflected in HIC studies.[27 28]

Implications for research and practice

The findings from the review suggest that PHRs may be particularly useful in health systems where little or no medical record keeping occurs at outpatient level, where record retrieval is difficult, or where multiple HCPs may be involved in a person's care for chronic conditions. This is especially important in LMICs as most have pluralistic health systems and no 'gate-keeping', so that patients often change their HCPs, or shop around, and receive care at both public and private facilities.[15 16 45] Patients' provision of a verbal medical history as the only means of transferring previous clinical

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information during HCP visits is, however, insufficient. In LMICs patients' communication of past medical history may be adversely affected by a lack of health literacy, inability to articulate the clinical procedures received, and a lack of confidence against a background of profound power differentials between patients and HCPs.[48-50]

A well-completed PHR may in principle improve medical information availability for HCPs. However, no trials, which assessed the availability of medical information for HCPs, were reported from LMICs. In Germany, Straßner et al[51] conducted a trial investigating the utility of PHRs for asylum seekers in reception centres for improving medical information availability for HCPs. The trial found that a PHR for asylum seekers increased the availability of health information for HCPs (aOR 4.22, 95% CI: 2.64 to 6.73), reduced missing information (aOR 0.89, 95% CI: 0.42 to 1.88), and reduced HCP dissatisfaction (aOR 0.43, 95% CI: 0.16 to 1.14) .[51] Previous HIC studies have shown that the use of PHRs for patients with diabetes and hypertension have led to decreases in HbA1c ($p<0.001$)[52-54], decrease in diastolic blood pressure ($p<0.05$) [52], and improvement in receiving foot examinations (OR: 1.68; 95% CI 1.12 to 2.50)[54], having physical activity or exercise advised (OR: 1.84; 95% CI 1.16 to 2.92) [54], and monitoring of weight, blood glucose and cholesterol[52]. Similar results may be achieved in LMICs for monitoring blood pressure/glucose/cholesterol, improving foot/eye examination if PHRs are well documented.

Patients, especially those with chronic diseases, usually carry the records they have to HCP visits, especially if they know HCPs will use them. [15 16 45 55] However, in the absence of an organised PHR, patients either bring very little information, such as a prescription alone (common in LMICs), or they bring all the previous pieces of paper that form their home held medical records, making it difficult for HCPs to find the relevant information.[16] Many chronic disease patients have multi-morbidities and often have numerous papers or clinic specific PHRs from the different clinics and hospitals they visit, but they often only carry notes from previous visits to the clinic they are going to visit.[28] Thus, the HCPs do not receive the information about medication and management plans produced by

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other doctors in other clinics or hospitals. Therefore, it is necessary to develop generic/universal and pragmatic PHRs and create awareness of the importance of using a single generic PHR for all healthcare visits, irrespective of the clinic/hospital, so that all HCPs have the necessary medical information available to make the best clinical decisions.

The WHO recommends that ideally a PHR should be used as complementary to a facility based medical record.[56] Prior reviews from HICs showed that, when HCPs had to enter information in facility based records (electronic or paper based) and PHRs, this multiple recording was an unwelcome burden.[57 58] However, documented discharge information for patients /families (e.g., summaries/letters) from HCPs has been found to reduce post-discharge complications[59] and re-admissions,[60] and to improve patient satisfaction[61] and health outcomes.[62] Arguably, it would more efficient and accurate if there were one record system accessible and useful to both patients and HCPs.

Many LMICs are progressing with the implementation of electronic health records, which may enable health systems to overcome difficulties in retrieving information from paper-based facility-based medical records.[63 64] Despite such advancements in technology and information technology capability in some middle-income countries, paper records will remain the prominent form of medical records and are unlikely to be completely replaced by electronic records at least for the near future. Furthermore, the issues of integrating electronic health records across different levels of care in health systems and public-private information exchange are often not addressed in existing health information management systems.[65] Therefore, PHRs having a minimum of data such as diagnosis, medication, lab results and follow-up will enable better information transfer in the health systems transitioning to electronic health records.

Strengths and limitations

Our systematic review has several strengths and limitations. The review protocol was registered at PROSPERO and we followed PRISMA guidelines in our systematic review. A comprehensive search

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strategy was employed to locate studies from LMICs. Furthermore, our search from the grey literature and discussion with experts in the field helped us to identify generic PHRs from Malawi,[66] Namibia[67] and South Africa.[68]However, these PHRs have not been evaluated systematically, and we have limited information on the usefulness of these records. However, our study also has some limitations. Given the small number of relevant articles, we found very few RCTs on which to base conclusions of robust experimental findings. Scarce data on generic PHRs from LMICs limit our ability to interpret the usefulness of these records in improving health outcomes of patients with NCDs.

CONCLUSION

Available data on the use of non-maternity PHR for handover communication between HCPs and HCPs with patients in LMICs are sparse. Existing studies on PHR in LMICs are of medium to low quality, and heterogeneous in terms of study design, population and context. Further, health outcome assessments after the introduction of PHRs are not available. The available studies show that patients value PHRs for their own use and are likely to carry them to HCPs. HCPs, in turn, consider that PHRs enhance documented information transfer from previous visits specifically when there is difficulty in retrieving past medical information. With LMICs progressing with electronic health record implementation, the role of paper based PHRs in overcoming issues of integration and for improving adherence to effective self-care needs re-evaluating. Future research should explore minimum information requirements for major conditions, the format and types of PHRs of most utility to patients and HCPs (electronic or paper-based), and barriers and facilitators to effective use of PHRs across all HCPs for any one patient. Further, the content and HCP training and protocols need to be developed and evaluated for the effective scale-up of interventions to promote the effective use of PHRs for continuity of care and patient self-care.

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Declaration of interest

We declare no competing interest.

Author contributions

JLM, PJ, SG and SMH conceived the article. JLM performed the literature search. JLM, BD, JP collected data from individual studies and interpreted the data. JLM wrote the first draft of the paper. SMH, SG, PJ, AL and CH revised the first and subsequent drafts. All authors contributed to interpretation of the findings and revised the manuscript for important intellectual content.

Data sharing

No additional data available

Fig 1. Legends

LMIC=Low and middle income countries, CVD=Cardiovascular disease, PHR=Patient-held health record, NCD=Non-communicable disease

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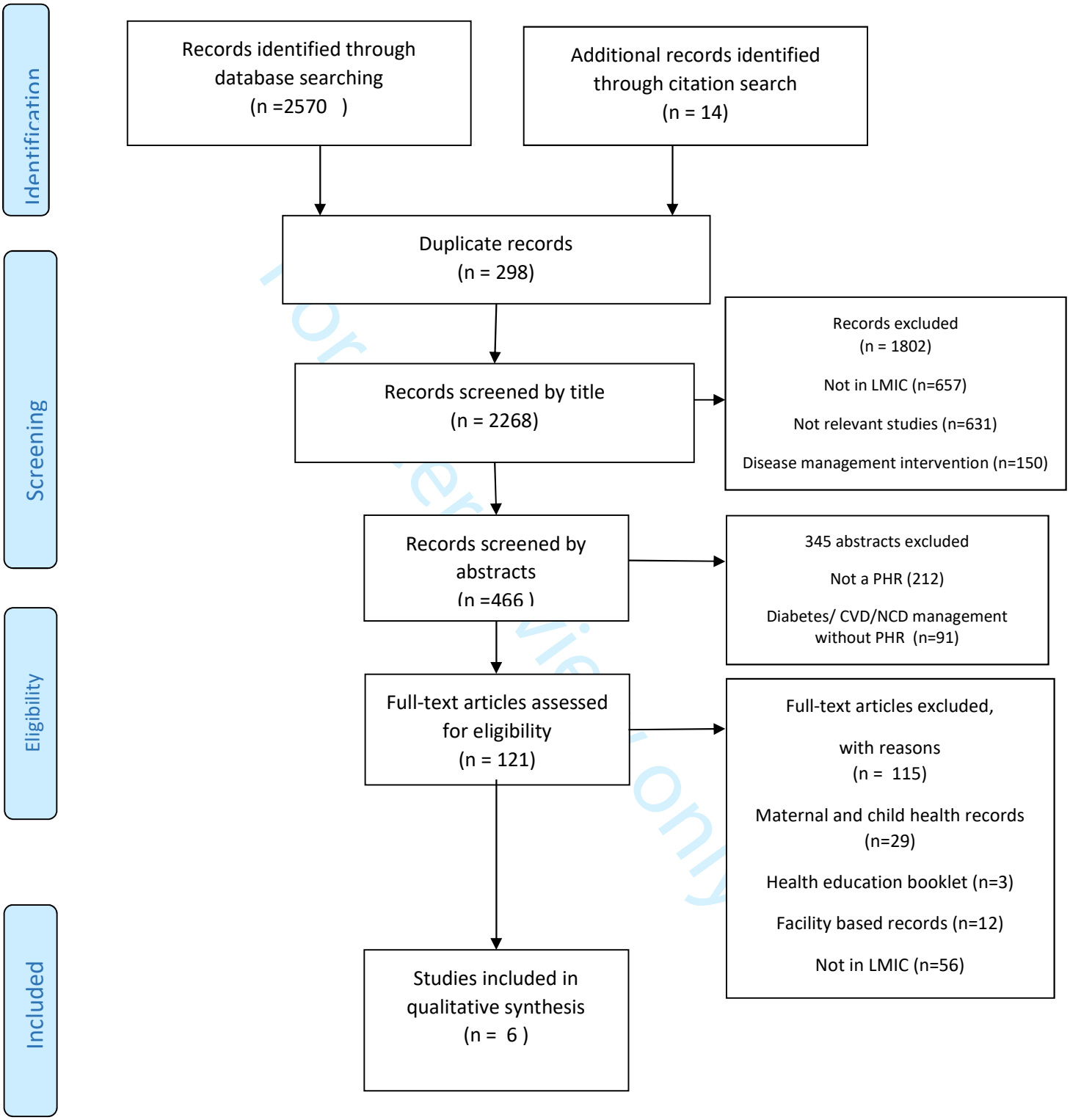


Fig 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram

SUPPLEMENTAL MATERIAL

S1.Search strategy in EMBASE

Patient-held records

1. (exp medical records/or exp medical records, problem-oriented/or exp medical records systems, computerized/or exp nursing records/or (((medical or health) and record*).mp. or (case note* or case record*).mp.) or (record* or book* or handbook* or card*).mp AND ((held adj3 (patient or parent or person or woman or man or family or consumer)).mp. or (carried adj3 (patient or parent or person or woman or man or family or client or consumer)).mp.)) OR ((log-book* or logbook*).mp. or home based record*.mp. or passport*.mp. or personal health record*.mp.)
2. ((patient-held or patient-held or home based or home-based or personal child or "mother and child" or "maternal and child" or mother-child) adj2 (record* or book* or handbook* or card*)).mp
3. ((patient access or personal or patient-held) adj2(record*or portal * or health record* or information* or prescription* or electronic record*)).mp
4. 1 or 2 or 3

Low-and Middle-income countries

5. exp developing country/ or exp medically underserved/ or developing countr$.mp. or medically underserved area$1.mp. or low income countr$.mp. or middle income country.mp. or low resource.mp. or resource poor.mp. or global.mp. or exp Africa/ or exp "South and Central America"/or exp asia/ or exp Caribbean islands/ or exp pacific islands/ or exp eastern Europe/ or exp Indian Ocean/or south america$1.mp. or Africa$1.mp. or Caribbean.mp. or central America$1.mp. or south America$1.mp. or eastern Europe$1.mp. or pacific island$.mp. or Indian ocean island$.mp. or asia.mp. or Afghan$.mp. or Bangladesh$1.mp. or Benin$.mp. or Burkina Faso.mp. or Burkinabe.mp. or Burundi$.mp. or Cambodia$1.mp. or Central African.mp. or Chad$.mp. or Comor$.mp. or Congo$.mp. or Eritrea$1.mp. or Ethiopia$1.mp. or Gambia$1.mp. or Guinea$1.mp. or Haiti$.mp. or Kenya$1.mp. or Korea$1.mp. or exp North Korea/ or Kyrgyz$.mp. or Liberia$1.mp. or Madagascar.mp. or Malagasy.mp. or Malawi$.mp. or mali$.mp. or mozambi$.mp. or Myanmar$.mp. or Nepal$.mp. or Niger$.mp. or Rwanda$1.mp. or Sierra Leone$.mp. or Somalia$1.mp. or Tajik$.mp. or Tanzania$1.mp. or Togo$.mp. or Uganda$1.mp. or Zimbabwe$.mp. or Angola$1.mp. or Armenia$1.mp. or Beliz$.mp. or Bhutan$.mp. or Bolivia$1.mp. or Cameroon$.mp. or Cape Verde$.mp. or Congo$.mp. or "Côte d'Ivoire".mp. or Ivory Coast.mp. or Ivorian.mp. or Djibouti.mp. or Egypt$.mp. or El Salvador.mp. or Salvadoran.mp. or Fiji$.mp. or Georgia$1.mp. or Ghana$.mp. or Guatemala$1.mp. or Guyan$.mp. or Hondura$.mp. or Indonesia$1.mp. or India$1.mp. or Iraq$1.mp. or Kiribati.mp. or Kosov$.mp. or Lao$.mp. or Lesotho.mp. or Marshall Islands.mp. or Marshallese.mp. or Mauritania$1.mp. or Micronesia$1.mp. or Moldov$.mp. or Mongolia$1.mp. or Morocc$.mp. or Nicaragua$1.mp. or Nigeria$1.mp. or Pakistan$1.mp. or Papua New Guinea$1.mp. or Paraguay$.mp. or Philippines.mp. or Filipino.mp. or Samoa$1.mp. or sao tome$.mp. or Senegal$.mp. or Solomon Island$.mp. or sri lanka$1.mp. or Sudan$.mp. or Swazi$.mp. or Syria$1.mp. or Timor$.mp. or

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Tonga$1.mp. or Turkmen$.mp. or Tuvalu$.mp. or Ukrain$.mp. or Uzbek$.mp. or Vanuat$1.mp. or Vietnam$.mp. or West Bank.mp. or Gaza.mp. or Yemen$.mp. or Zambia$1.mp. or Albania$1.mp. or Algeria$1.mp. or "Antigua and Barbuda".mp. or antiguan.mp. or barbudan.mp. or Azerbaijan$1.mp. or Belarus$.mp. or Bosnia$1.mp. or Botswana.mp. or Brazil$.mp. or Bulgaria$1.mp. or Chile$.mp. or China.mp. or Chinese.mp. or Colombia$1.mp. or Costa Rica$1.mp. or Cuba$1.mp. or Dominica$1.mp. or Ecuador$.mp. or Gabon$.mp. or Grenad$.mp. or Iran$.mp. or Jamaica$1.mp. or Jordan$.mp. or Kazakhstan$1.mp. or Latvia$1.mp. or Leban$.mp. or Libya$1.mp. or Lithuania$1.mp. or Macedonia$1.mp. or Malaysia$1.mp. or Maldiv$.mp. or mauriti$.mp. or Mexic$.mp. or Montenegr$.mp. or Namibia$1.mp. or Palau$.mp. or Panama$.mp. or Peru$.mp. or Romania$1.mp. or Russia$1.mp. or Serbia$1.mp. or Seychell$.mp. or South Africa$1.mp. or Saint Kitts.mp. or Saint Lucia.mp. or Saint Vincent.mp. or Suriname$1.mp. or Thai$.mp. or Tunisia$1.mp. or Turk$.mp. or Uruguay$.mp. or Venezuala$1.mp

6. 4 and 5

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S2. Table summaries of MMAT quality appraisal assessments for included studies

Author, year, LMIC	Type of study	4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)
Van der Hoek [23], 1995, Zambia	Quantitative descriptive	Can't tell	Can't tell	Yes	No	Yes	2
Henbest[24], 1995, Lesotho	Quantitative descriptive	Yes	Yes	Can't tell	Yes	Yes	4
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
Norden[25],2004 South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Kerry[26] , 2005, South Africa	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
Chen[27] et al, 2014,China	Qualitative	Yes	Yes	Yes	Yes	Can't tell	4
		4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?	Score (Out of 5)

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Ibrahim[28] et al, 2019, Mongolia	Quantitative descriptive	Yes	Yes	Yes	Yes	Yes	5 (Has description of non-responders and confounders)
		1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?	
	Qualitative	Yes	Yes	Can't tell	Yes	Can't tell	3 (Very little detail on coding and data analysis)
		5.1. Is there an adequate rationale for using a mixed methods design to address the research question?	5.2. Are the different components of the study effectively integrated to answer the research question?	5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?	
	Mixed methods	Yes	Can't tell	Yes	Can't tell	Yes	3(Integration may have occurred in the interpretation of results-It is unclear, cannot tell if there is divergent findings)

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1, Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5,7
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7,8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file, S1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	10,11

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12,13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13 and Table 3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 and Supplementary file, S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	18-20
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Na
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20

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Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	23
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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